

From water to dielectric fluids: Pool boiling research under pressure and subcooling for next-generation cooling

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ABSTRACT

The rapid growth of data-intensive technologies in renewable energy systems has increased the demand for efficient thermal management of high-performance power electronics in data centers and battery energy storage systems (ESS). Dielectric fluids are attractive working fluids for immersion cooling owing to their electrical insulation, chemical stability, and low boiling temperatures. However, their relatively low critical heat flux (CHF) and heat transfer coefficient (HTC) compared with water remain major limitations. While prior studies have largely focused on surface engineering to enhance boiling performance, this review instead synthesizes existing experimental data and predictive correlations for CHF and HTC of dielectric fluids, with particular emphasis on the effects of system pressure and liquid subcooling. Owing to their distinct thermophysical properties, such as low surface tension and latent heat, dielectric fluids exhibit fundamentally different dimensionless parameters and sensitivities to pressure and subcooling compared with water, making universal boiling correlations inapplicable. By critically reviewing correlation performance across operating conditions, this work clarifies their applicability and limitations for dielectric fluids. A qualitative techno-economic discussion is also provided to link boiling performance trends to potential system-level energy and cost benefits. Overall, this review highlights the potential of pressure- and subcooling-based strategies for improving boiling performance in dielectric fluids and identifies key knowledge gaps that must be addressed to establish robust and generalizable predictive frameworks.

Nomenclature

ESS	Energy Storage System
CHF	Critical Heat Flux [W/m^2]
HTC	Heat Transfer Coefficient [$\text{W}/\text{m}^2\text{K}$]
BHT	Boiling Heat Transfer
ONB	Onset of Nucleate Boiling
AI	Artificial Intelligence
ANN	Artificial Neural Networks
R ²	Coefficient of determination
MSE	Mean Square Error
XAI	Explainable Artificial Intelligence
GA	Generic Algorithm
HSN	Heterogeneous Spontaneous Nucleation
HI	Hydrodynamic Instability
GWP	Global Warming Potential
ODP	Ozone Depletion Potential

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Pr	Prandtl number
Bo	Bond number
We	Weber number
Ja	Jakob number
Pe	Peclet number
Pc	Critical pressure [Pa]
Pred	Reduced pressure (P/Pc)
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error [%]
HTCMC	High-Thermal-Conductivity Metal-Ceramic Composite
TEA	Techno-Economic Analysis
LCA	Life-Cycle Assessment
PW	Present Worth
CPW	Cumulative Present Worth
FCI	Fixed Capital Investment
PEC	Purchased Equipment Cost

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C_{energy}	Energy cost
C_{carbon}	Carbon emission cost
C_{OM}	Operation and maintenance cost
C_{R}	Replacement cost
CRF	Capital Recovery Factor
QS	Total electrical power consumption [kW]
Δ	Difference
ρ	Density (kg/m^3)
σ	Surface tension (N/m)
ν	Kinematic viscosity (m^2/s)
α	Thermal diffusivity (m^2/s)
Subscripts	
sat	Saturation
w	Wall
super	Superheat
sub	Subcooling
OPEX	Operational expenditures

1. Introduction

Boiling heat transfer (BHT) leverages liquid–vapor phase change to achieve high heat flux dissipation at relatively small temperature differences [1–5]. Repeated bubble formation, growth, and departure at nucleation sites enhance convective transport, making BHT particularly effective for high rate heat transfer. Among its configurations, pool boiling remains one of the simplest and most cost-effective. By immersing heated components in a quiescent liquid pool, large amounts of heat can be removed without complex auxiliary equipment, offering advantages in both efficiency and system simplicity. This principle underpins a wide range of industrial and scientific applications, from refrigeration [6] and nuclear systems [7,8] to aerospace [9] and electronics cooling [10]. More recently, pool boiling has also gained importance in renewable energy contexts [11–15], such as concentrated solar power [16], geothermal energy [17], and fuel-cell thermal stabilization [18].

The resurgence of interest in pool boiling arises from its relevance to compact, high efficiency cooling in high performance and high energy density devices [19]. Here, dielectric fluids—including fluorocarbons and hydrofluoroethers—have attracted attention due to their electrical insulation, chemical stability, and compatibility with sensitive electronic components [20]. Their low boiling points and favorable dielectric properties make them suitable for two-phase cooling in data centers [21], satellites [22], battery energy storage systems (ESS) [23], and other high power density environments involving electrical currents. In this regard, accurate prediction of the boiling behavior of dielectric fluids is therefore essential for the design of next-generation, sustainable cooling solutions.

This need is further amplified by the rapid expansion of data-intensive infrastructures associated with renewable energy systems, such as smart grids and digital-twin platforms, which demand real-time monitoring and high-speed data processing [24,25]. The resulting growth in high-performance computing hardware has substantially increased thermal loads in data centers [26]. In parallel, the intermittency of renewable energy sources has driven widespread deployment of battery ESS to buffer power fluctuations and stabilize electrical grids [27]. As these systems are operated under strong electrical constraints, advanced thermal-management strategies based on dielectric fluids are increasingly indispensable for improving both energy efficiency and operational sustainability.

With this momentum, several reviews have summarized boiling studies of dielectric fluids in recent years [28–33]. To clarify the distinct contribution of the present study, Table 1 summarizes the scope and limitations of these recent review articles. As shown in the table, the majority of prior reviews have focused on heat transfer enhancement techniques via surface modification [28,30] or the general mechanisms

Table 1

Summary of recent review articles on pool boiling and comparison with the present study.

Author (Year)	Main Scope & Focus	Limitations/Gaps addressed by present study
Leong et al. (2017) [28]	Dielectric fluids on enhanced surfaces (porous foams, micro-fins, etc.).	Focuses on performance augmentation via surface modifications; does not address fundamental correlation accuracy for plain surfaces.
Liang & Mudawar (2018) [29]	Comprehensive review of CHF mechanisms and theoretical models for general fluids.	Provides a universal theoretical framework (heavily based on water) rather than a practical benchmarking of correlations specifically for dielectric fluids.
Sajjad et al. (2020) [30]	Enhancement mechanisms for dielectric and highly wetting liquids on micro/nano-engineered surfaces.	Analyzes how surface structures improve boiling; lacks a baseline assessment of predictive correlations for plain surfaces under varying pressure.
Emir et al. (2022) [31]	Parametric effects (pressure, subcooling, orientation) on CHF and HTC trends.	Reviews phenomenological trends (e.g., effect of pressure increase) but does not perform a statistical evaluation to identify the most accurate correlations for dielectric fluids.
Volodin et al. (2025) [32]	Reviews boiling heat transfer enhancement of dielectric fluids specifically through various surface modification techniques.	Catalogs effective modification methods based on fluid types and operating conditions; lacks validation of the accuracy and applicability of correlations.
Pereira et al. (2026) [33]	Comprehensive assessment of pool boiling enhancement strategies, including active methods (e.g., magnetic fields, additives) alongside surface modifications.	Does not provide a comparative analysis of effectiveness relative to varying fluid properties and lacks a correlation-based performance evaluation.
Present Study	Dielectric fluids on plain surfaces under pressure & subcooling.	Fills the gap by systematically evaluating the accuracy of CHF/HTC correlations to identify the best design tools for high-pressure and subcooled immersion cooling.

of the critical heat flux (CHF) [29]. Recently, the effects of parametric trends [31], surface modifications [32], and other active methods [33], have also been reviewed. However, a rigorous quantitative and comparative assessment of correlations specifically for dielectric fluids is still insufficient. This study bridges this gap by systematically evaluating the predictive accuracy of CHF and heat transfer coefficient (HTC) correlations for dielectric fluids on plain surfaces under the combined effects of system pressure and liquid subcooling.

Despite these previous efforts, a critical knowledge gap persists regarding the fundamental baseline behavior of dielectric fluids. Although pressure-dependent correlations for CHF are well established for water, systematic cross-fluid evaluations for dielectric liquids are limited, and controlled datasets under subcooled conditions remain especially sparse. Historically, pool boiling research centered on using water under saturated conditions [34]. By contrast, emerging applications in thermal management systems increasingly involve dielectric liquids under non-saturated, transient, or spatially localized conditions [35]. Extending correlations from water to such cases require careful consideration of subcooling, yet the supporting database is insufficient. Furthermore, even when datasets exist, their interpretation is complicated by the coupled influence of pressure, subcooling, fluid properties, and surface characteristics, making it difficult to isolate the role of any single parameter. This complexity undermines cross-study comparisons and underscores the need for carefully designed experiments.

This review addresses that gap by systematically assessing CHF and HTC behavior in dielectric fluids across operating conditions, with emphasis on the coupled effects of system pressure and subcooling. To

ensure consistency, the analysis is restricted to plain (untreated) surfaces and rigorously controlled experiments. We compared experimental data against established CHF models, highlighted where predictions are accurate or not, and identified the limitations of extending water-based correlations to dielectric fluids. Based on this critical appraisal, we outlined research priorities to advance predictive capability and deepened the understanding of pool boiling mechanisms, thereby supporting the development of sustainable and reliable thermal management strategies.

Beyond thermal performance metrics, improvements in CHF and HTC also have important system-level and economic implications in two-phase immersion cooling. Higher boiling limits and enhanced heat transfer relax constraints on operating margins, circulation requirements, and heat rejection capacity, which can reduce auxiliary power demand and improve operational flexibility. Accordingly, this review includes a brief techno-economic analysis to highlight the potential energy-efficiency and cost benefits associated with improved boiling performance.

The remainder of this review is organized as follows. Section 2 introduces the fundamentals of pool boiling heat transfer. Section 3 summarizes the boiling behavior of dielectric fluids. Section 4 reviews commonly used CHF and HTC correlations. Sections 5 and 6 examine the effects of pressure and subcooling on CHF and HTC, respectively. Finally, Section 7 discusses the qualitative techno-economic implications of pressure and subcooling effects in two-phase immersion cooling systems.

2. Fundamentals of pool boiling heat transfer

This section reviews the fundamental mechanisms and characteristics of pool boiling heat transfer. The discussion begins with an overview of boiling regimes and the representative boiling curve, followed by a description of the key performance parameters—CHF and HTC—and the influence of operating parameters such as system pressure and liquid subcooling. Enhancement strategies for improving CHF and HTC are also introduced, including both active and passive approaches. Finally, the section narrows its focus toward the classical empirical and semi-empirical correlations that form the basis for performance prediction, providing the necessary background for the subsequent analysis of dielectric fluids in Section 3.

2.1. Boiling regime and the boiling curve

Boiling heat transfer is governed by distinct thermal regimes that can be effectively characterized using a representative performance curve, such as the boiling curve [36]. The boiling curve, illustrated in Fig. 1, describes the relationship between wall heat flux and wall superheat, defined as the temperature difference between the heated surface and the saturation temperature of the general fluid. As wall superheat increases, the heat transfer mechanism transitions sequentially through (i) natural convection, (ii) nucleate boiling, (iii) transition boiling, and (iv) film boiling. Each regime reflects a fundamental shift in interfacial behavior and thermal transport processes.

The nucleate boiling regime, marked by active bubble nucleation, growth, and departure, provides the highest HTC values due to enhanced heat exchange and vigorous fluid mixing. In contrast, film boiling, characterized by the formation of a stable vapor layer over the surface, significantly deteriorates heat transfer performance due to the low thermal conductivity of vapor. The key transition points between these regimes are: the onset of nucleate boiling (ONB), at which vapor bubbles first begin to form on the heated surface; the CHF, which marks the maximum heat flux that can be sustained before the onset of boiling crisis and the transition to film boiling; and the Leidenfrost point, where the stable vapor film collapses during surface cooling, leading to re-establishment of nucleate boiling.

While water is the most extensively studied fluid in pool boiling research, dielectric fluids—widely used in systems involving electrical currents—exhibit the same boiling regimes but with notable differences in transition points and slope of the boiling curve [20,30]. These differences arise from their lower surface tension, reduced latent heat, and lower thermal conductivity compared to water, which affect bubble dynamics and heat transfer mechanisms. The implications of these differences will be discussed in detail in Section 3.

In addition to these shifts in boiling regime transitions, dielectric fluids are also prone to an instability phenomenon known as boiling excursion, which is rarely observed in pool boiling of water under comparable conditions [37]. Boiling excursion is characterized by an abrupt and disproportionate increase in wall superheat in response to a relatively small change in heat flux or operating conditions, leading to a sudden departure from the stable nucleate boiling branch of the boiling curve. This behavior has been frequently reported for dielectric fluids, such as FC-72 and HFE-7100, where low surface tension, low latent heat of vaporization, and weak liquid replenishment near the heated surface promote thermally unstable boiling behavior [30,37–46]. In many

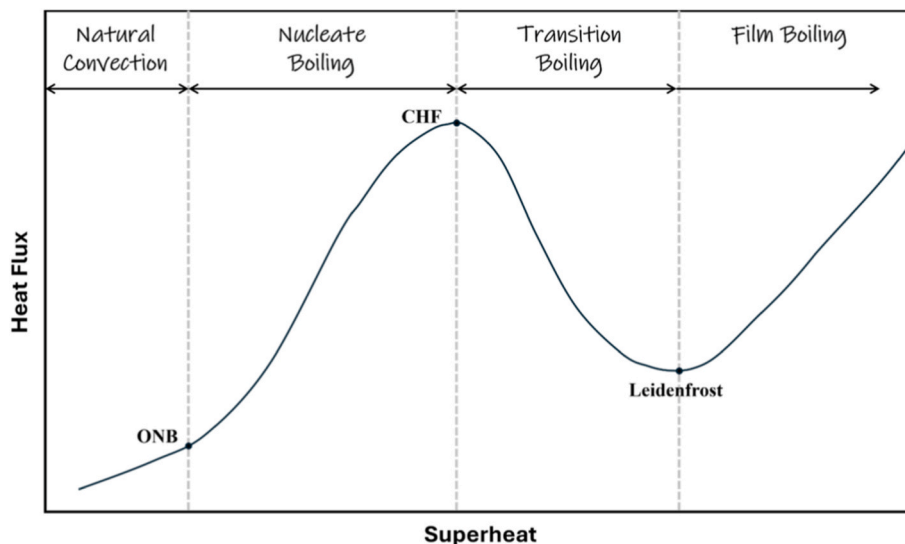


Fig. 1. Typical pool boiling curve.

applications of interest, such thermal excursions may remain within acceptable temperature limits and may only marginally reduce the overall attractiveness of immersion cooling [37].

However, from a device reliability perspective, boiling excursion raises important concerns [37,38]. A rapid and localized rise in wall temperature can induce steep temperature gradients across the chip surface, which may generate significant thermal stresses. Repeated exposure to such thermal gradients can promote crack initiation and propagation within the chip or its packaging, ultimately shortening component lifetime. In addition, if the temperature overshoot associated with boiling incipience and excursion cannot be adequately controlled, maintaining chip temperatures within the required operating range throughout the system lifetime becomes increasingly challenging. These issues, together with the roles of thermophysical properties, system pressure, and liquid subcooling in boiling excursion behavior, are discussed in detail in Section 3.

2.2. Key parameters in boiling phenomena: CHF and HTC

Pool boiling heat transfer performance is typically characterized by two essential parameters: the CHF and the HTC. Both parameters play vital roles in the design and operation of thermal management systems across a wide range of industrial applications [47].

The **critical heat flux (CHF)** is the maximum heat flux a heated surface can sustain before transitioning from nucleate boiling to film boiling—a regime marked by a sharp drop in heat transfer efficiency and a rapid rise in surface temperature [36]. Once CHF is exceeded, formation of the vapor blanket severely impedes heat transfer, potentially causing significant performance degradation or even catastrophic system failure. Consequently, CHF is widely regarded as a fundamental safety limit in boiling based systems.

The **heat transfer coefficient (HTC)**, defined as the ratio of heat flux to wall superheat, as expressed as: $h = q'' / \Delta T_{sat} = q'' / (T_w - T_{sat})$. It serves as a key indicator of thermal performance [36]. A higher HTC reflects more effective heat removal at a given temperature gradient, which is crucial for compact and high performance cooling designs.

Improving CHF enhances safety margins, whereas increasing HTC directly improves performance. Both are indispensable for advancing boiling based thermal management. Consequently, decades of research have been devoted to enhancing these parameters, using strategies broadly classified into **active** and **passive** approaches [4,48–52].

- **Active methods** involve the use of external inputs, such as mechanical agitation, surface vibration, or electric/magnetic fields, to influence fluid motion.
- **Passive methods**, in contrast, require no external energy and include surface morphology modification (e.g., roughening, porous coatings, artificial cavities) and adjustment of operating conditions (e.g., pressure, subcooling).

2.3. AI based prediction for CHF and HTC

Recent studies have focused on employing artificial intelligence (AI) to predict the CHF and HTC of boiling heat transfer. AI based prediction methods enable more time-saving and reliable estimation of the CHF and HTC across various parameters, facilitating efficient optimization of surface designs and operating conditions. Some recently published literature integrating AI with pool boiling heat transfer have employed machine learning, which is a subset of AI that allows the system to learn from data and improve its performance. A review about the application of machine learning on boiling heat transfer suggested through statistical analysis, that the number of related articles had increased exponentially in the past few years [53].

Algorithms based on artificial neural networks (ANN) are currently the most commonly used in machine learning for fitting non linear data of boiling heat transfer, due to their high accuracy [54]. ANNs are

composed of multiple layers of connected processing units that capture complex, nonlinear relationships between input and output parameters. Machine learning is used to construct a functional relationship between the input and output spaces. The dataset is separated into training, testing and validation subsets. A function between the inputs and outputs, which generalizes the training subset, is derived. The effectiveness of the model is evaluated based on its predictive accuracy.

Al-Hindawi et al. [55] developed an ANN model to predict the pool boiling HTC of nanofluids based on a dataset of 1342 experimental data from the previous studies. The effects of parameters such as heat flux, saturation pressure, nanoparticle and fluid thermal conductivity, nanoparticle concentration and size, on the HTC were investigated across various operating conditions. The ANN model with 19 neurons in one hidden layer, which used tansig and purelin transfer functions, was evaluated to perform with the highest accuracy, with a coefficient of determination (R2) of 0.9948 and overall mean square error (MSE) of 0.01529.

Sajjad et al. [56] improved the prediction accuracy of pool boiling CHF by combining physics-informed principles with interpretable AI frameworks. A deep neural network (DNN) model was developed by using a dataset from 42 pool boiling studies. Applying this model resulted in enhanced prediction accuracy of CHF, exhibiting an R2 value of 0.97. The key parameters of boiling heat transfer such as surface inclination, thermal conductivity, saturation temperature, wall superheat, and surface roughness were incorporated into the prediction model. The model also applied explainable artificial intelligence (XAI) to analyze the prediction results, where surface inclination was identified as the dominant parameter.

Sajjad et al. [57] identified the optimal operational and surface morphological parameters of heat sinks in two-phase thermal systems, by using the DNN model developed in the previous study. To predict the optimal design and operational parameters, an optimized genetic algorithm (GA) model was proposed. The GA model showed reduced inclination angles with increased target CHF, which is aligned with experimental observations of reduced vapor removal at higher angles. Surface roughness and liquid saturation temperature data were also consistent with experimental boiling mechanisms, which show that the model effectively predicts fundamental thermofluidic phenomena despite being data-driven.

2.4. Influence of pressure, subcooling on boiling heat transfer

The CHF and HTC in pool boiling are strongly influenced by system pressure and liquid subcooling, which are generally classified as passive control parameters that alter both the thermophysical properties of the liquid and the hydrodynamics of vapor bubbles [36]. These parameters govern key boiling phenomena, including the onset of nucleate boiling, bubble growth and detachment, coalescence, and vapor blanket formation, all of which directly affect boiling heat transfer performance and stability.

In the following subsections, the individual roles of system pressure and liquid subcooling are examined in detail, with particular emphasis on their distinct physical mechanisms and their combined influence on boiling performance. The overall trends in boiling characteristics are summarized schematically, and the corresponding effects on bubble dynamics and CHF/HTC behavior are illustrated in Figs. 2, 3 and 6. Together, these figures provide a unified visual framework for interpreting the experimental observations and theoretical analyses reported in the literature.

• Effect of System Pressure

System pressure exerts a strong influence on both the thermophysical properties of the liquid and the hydrodynamics of vapor bubble formation. At elevated pressures, the critical radius for bubble nucleation decreases, leading to the formation of smaller vapor bubbles that detach

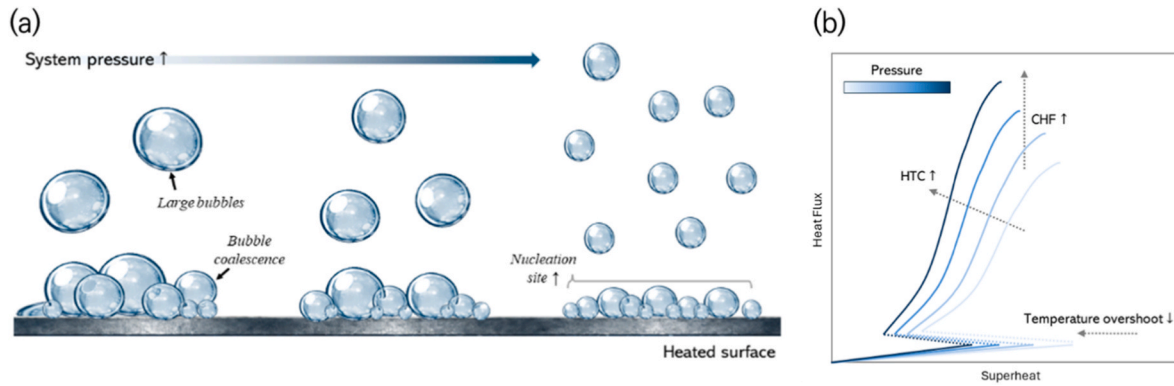


Fig. 2. Schematic representation of the effect of pressure on pool boiling system: (a) Modification of bubble dynamics, showing the reduction in bubble departure diameter and the increase in active nucleation site density with increasing pressure; (b) Shift in the boiling curve, illustrating the increase in CHF and HTC, along with the suppression of temperature overshoot at elevated pressures.

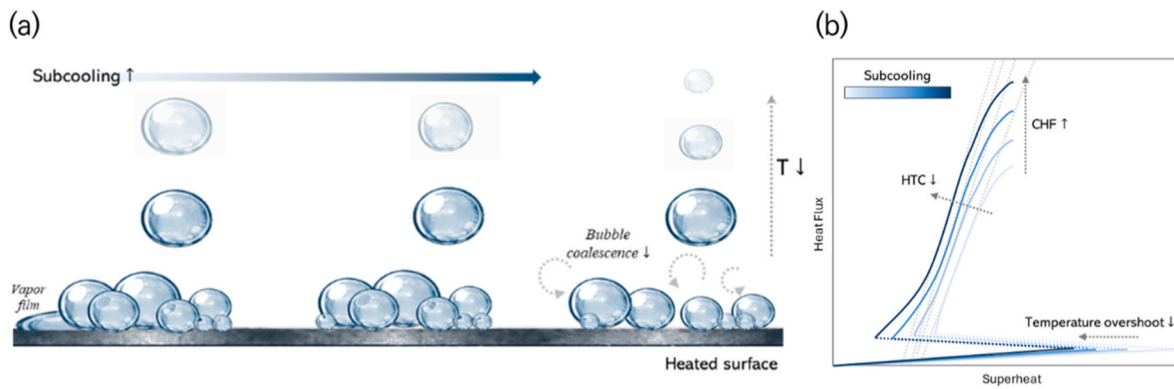


Fig. 3. Schematic representation of the effect of liquid subcooling on pool boiling system: (a) Modification of bubble dynamics, illustrating how enhanced convection and vapor condensation suppress vapor film formation and improve stability; (b) Shift in the boiling curve, showing the delayed ONB, the mitigation of temperature overshoot, and the significant enhancement of CHF.

from the heating surface at a higher frequency [58,59]. The increase in vapor density reduces buoyancy forces acting on the bubbles, making their behavior more dependent on surface tension [60]. In this regime, surface tension plays a greater role in balancing cohesive forces within the liquid against adhesive forces between the liquid and the solid surface, promoting more uniform and stable nucleation across multiple sites. These effects generally enhance CHF by facilitating more frequent liquid–surface rewetting and limiting the size of vapor structures [58–60].

From a thermodynamic perspective, increasing system pressure raises the boiling point of the liquid. This is because phase change occurs when the vapor pressure equals the external pressure, and at higher pressures, a greater vapor pressure—and thus higher liquid temperature—is required to initiate boiling [61,62]. While this allows operation at higher bulk liquid temperatures without phase change, it can also delay the onset of nucleate boiling, reducing the fraction of heat transfer accomplished via latent heat. At very high pressures, particularly near the critical point, smaller bubbles tend to coalesce rapidly into larger vapor structures. This can trigger premature transition from nucleate to film boiling, in which a continuous vapor layer forms over the surface, acting as a thermal insulator and drastically reducing heat transfer performance [63]. Therefore, while moderate pressure increases are beneficial for CHF enhancement, excessive pressures can be detrimental.

Numerous experimental studies have investigated the influence of pressure on boiling behavior. Sakashita et al. [64] observed that as pressure increased from 1 to 5 MPa in saturated water, the size of the primary bubbles decreased, while their detachment frequency remained mostly unchanged. The CHF increased with pressure, suggesting

improved thermal stability, although this effect was not directly linked to bubble departure frequency. Subsequent studies have further demonstrated that the increase in CHF with pressure is more closely associated with the activation of a larger number of nucleation sites rather than with changes in bubble departure frequency [65]. Elevated pressure enhances nucleation site density by reducing the critical nucleation radius and modifying surface wettability, leading to more uniformly distributed boiling activity and improved liquid–surface rewetting, which collectively contribute to enhanced thermal stability. Dahariya et al. [66] further showed that changes in the thermophysical properties of water due to pressure significantly altered bubble hydrodynamics and the development of the thermal boundary layer.

In dielectric fluids, similar trends have been reported, although the magnitudes and mechanisms can differ due to their distinct properties. Rainey et al. [67] demonstrated that increasing pressure from 30 to 150 kPa in FC-72 pool boiling delayed the onset of dryout and increased CHF. Guan et al. [68] observed that CHF consistently increased with pressure in pentane, hexane, and FC-72, with hexane showing the greatest enhancement. In addition, pressure effects are strongly coupled with surface conditions. Dahariya et al. [69] reported that sintered copper wick structures under elevated pressures exhibited significantly enhanced CHF and HTC compared to plain surfaces, due to increased nucleation activity and bubble detachment.

In addition to CHF enhancement, increasing pressure has been shown to mitigate temperature overshoot and boiling excursion phenomena, which are characteristic challenges in the pool boiling of dielectric fluids. Numerous studies have reported abrupt wall-temperature excursions near boiling incipience in nonpolar dielectric

fluids such as FC-72 and HFE-7100 [30,37–46]. These instabilities are often attributed to the lack of pre-existing vapor embryos and the sudden activation of nucleation sites. The introduction of microstructured or porous surfaces has been well known to alleviate or even eliminate boiling excursion by providing stable vapor entrapment sites that serve as nucleation triggers [38–41]. Importantly, experimental evidence indicates that even on plain surfaces, increasing system pressure can significantly suppress temperature overshoot and boiling excursion [42]. This behavior can be consistently explained by the pressure-induced activation of a larger number of nucleation sites, which stabilizes the boiling process by distributing vapor generation more uniformly and preventing localized overheating.

In summary, the elevation of system pressure exerts a multifaceted influence on pool boiling performance. It enhances thermal transport by modifying fluid properties and bubble hydrodynamics, leading to a simultaneous increase in both the CHF and HTC. Furthermore, pressure plays a critical role in stabilizing the boiling incipience of highly wetting dielectric fluids by mitigating temperature overshoot. To visually summarize these trends, Fig. 2 schematically illustrates the pressure-dependent modifications in bubble dynamics and the resulting shifts in the boiling curve. As depicted in Fig. 2 (a), increasing pressure leads to a reduction in bubble departure diameter and a significant increase in active nucleation site density. These microscale changes directly enhance thermal transport, manifesting as a leftward shift in the boiling curve as shown in Fig. 2 (b). This shift indicates higher heat transfer efficiency, an elevated CHF limit, and the suppression of the temperature overshoot (boiling excursion) typical of dielectric fluids.

• Effect of Liquid Subcooling

Liquid subcooling (the temperature difference between the saturation temperature and the actual bulk liquid temperature) significantly affects boiling initiation, bubble dynamics, and CHF. Increased subcooling enhances the thermal driving force by enlarging the temperature gradient between the heated wall and the liquid bulk. In subcooled boiling, a greater wall superheat is required to raise the near-surface liquid temperature to saturation, delaying the onset of bubble nucleation and resulting in smaller, more frequent bubbles [70,71]. This delay shifts the boiling curve toward higher wall superheats and allows more thermal energy to be stored in the liquid before phase change occurs. Consequently, CHF is increased because the formation of an insulating vapor film is suppressed and liquid–surface contact is maintained for longer durations [72,73].

Subcooling is especially advantageous in high heat flux applications, where stable boiling is essential for avoiding localized dryout and ensuring uniform heat removal. By keeping the near-wall liquid in a metastable liquid state, subcooling prevents premature vapor blanket formation and maintains low thermal resistance between the surface and coolant. However, excessive subcooling may be counterproductive, as it requires disproportionately high wall superheats to initiate boiling, which can reduce the system's overall thermal efficiency [74]. Therefore, an optimal subcooling level should be identified to maximize CHF gains while maintaining efficient operation.

Experimental results support this behavior. Jun et al. [75] reported that CHF increased by approximately 60 kW/m² for every 10 K increase in subcooling on both plain and microporous copper surfaces, while HTC showed minimal variation. In dielectric fluids, Rainey et al. [67] observed similar CHF enhancements with increasing subcooling in FC-72, particularly on structured surfaces. Guan et al. [68] showed that subcooling decreased wall superheat for a given heat flux, leading to an increase in HTC in the nucleate boiling regime. For dielectric fluids, the lower latent heat and thermal conductivity may moderate the potential CHF enhancement by subcooling, especially at high heat flux.

Liquid subcooling enhances thermal stability during boiling incipience by strengthening single-phase convective heat transfer prior to nucleation. As the degree of subcooling increases, improved convective

heat removal allows the wall temperature to rise more gradually with increasing heat flux, thereby reducing the magnitude of the wall superheat overshoot typically observed at boiling onset [39].

In addition, subcooling shifts the onset of nucleate boiling to higher heat flux levels, narrowing the temperature gap between the terminal single-phase convection branch and the nucleate boiling branch of the boiling curve. This smoother transition mitigates abrupt wall temperature drops and associated instabilities, ultimately reducing the boiling excursion in dielectric fluid pool boiling systems [39–41].

In summary, liquid subcooling serves as a critical stabilizing mechanism in pool boiling systems. By enhancing single-phase convective heat transfer prior to nucleation, subcooling improves thermal stability during boiling incipience and effectively suppresses the formation of an insulating vapor film. This mechanism is instrumental in mitigating the large wall temperature overshoot often observed in highly wetting fluids. To visually summarize these trends, Fig. 3 schematically illustrates the subcooling-induced stabilization of bubble dynamics and the resulting shifts in the boiling curve. As depicted in Fig. 3 (a), the enhanced convective heat removal and rapid vapor condensation in the subcooled bulk liquid prevent the coalescence of bubbles into vapor patches, thereby maintaining stable liquid–surface contact. This directly translates into the boiling curve behavior shown in Fig. 3 (b): although the ONB is delayed to higher wall superheats, the transition from single-phase convection to boiling is smoother—indicating a suppression of overshoot—and the CHF limit is significantly elevated.

• Synergistic Effect of Pressure and Liquid Subcooling

System pressure and liquid subcooling can act synergistically in enhancing boiling heat transfer. Pressure primarily influences bubble nucleation size, detachment behavior, and surface wetting, while subcooling strengthens the thermal driving force and delays the formation of insulating vapor films. When optimized simultaneously, they can substantially increase CHF and improve boiling stability. However, only a limited number of studies have been conducted in this context.

Watwe et al. [76] experimentally investigated the combined influence of pressure and subcooling on FC-72 boiling in PPGA chip packages. While pressure increase elevated the saturated CHF, the enhancement diminished beyond a certain threshold, and the effect of subcooling on CHF rapidly weakened with increasing pressure. Importantly, the two effects were not additive, suggesting complex coupling between nucleation and liquid supply mechanisms. Rainey et al. [77] also studied pressure, subcooling, and non-condensable gas effects on both microporous and plain surfaces. Their results confirmed that pressure and subcooling independently increased CHF, but the magnitude of subcooling induced enhancement was more pronounced on microporous surfaces compared to plain ones, again without clear evidence of a synergistic interaction in FC-72.

More detailed insights were provided by Sakurai and Fukuda [78], who systematically measured subcooled pool boiling CHF of water, liquid nitrogen, and liquid helium on a 1.2 mm diameter platinum cylinder under varying pressures. Their work revealed a nonlinear dependence of CHF on subcooling, which could be categorized into three distinct regimes: (i) at low subcooling, CHF increased with pressure, consistent with the hydrodynamic instability mechanism; (ii) at high subcooling, CHF also increased with subcooling but exhibited to be nearly independent of pressure, attributed to heterogeneous spontaneous nucleation (HSN); and (iii) in an intermediate regime, CHF exhibited a decrease from a maximum to a minimum, although in some cases this transition was absent and only hydrodynamic instability (HI) or HSN driven trends persisted. This study emphasized that the interplay of pressure and subcooling cannot be captured by simple linear models and highlighted the role of HSN as a distinct mechanism in high subcooling conditions.

3. Boiling behavior of dielectric fluids

3.1. Thermophysical properties of dielectric fluids

The type of working fluid used in a pool boiling system significantly influences heat transfer performance and bubble dynamics [79], and in this review the primary focus is on dielectric fluids. To provide a comprehensive understanding of their thermophysical characteristics, this section compares the key properties of commonly used dielectric fluids, such as fluorocarbons (e.g., FC-72) and hydrofluoroethers (e.g., HFE-7100, Novec-649), against those of water, which serves as the reference fluid. The representative properties are organized in Table 2.

• Water

Water is widely used in pool boiling experiments and applications due to its excellent thermophysical properties and high heat transfer performance. Its high specific heat capacity allows substantial energy absorption as latent heat during phase change, enabling efficient thermal energy storage with minimal temperature fluctuation—an advantage in applications sensitive to temperature variations. In addition, water's high thermal conductivity supports efficient heat transfer from the heated surface to the surrounding liquid. Other advantages include abundance, low cost, environmental friendliness, and non-toxicity.

However, water has significant limitations in electrically sensitive applications. Due to its polarity, water readily dissolves impurities, and the resulting ions act as charge carriers under an electric field, making it electrically conductive. While deionization can temporarily reduce conductivity, deionized water quickly reabsorbs ions from its surroundings (e.g., from pipes and chambers), negating the effect. This intrinsic tendency toward ion uptake limits its suitability in applications requiring electrical insulation.

• Dielectric Fluids

Dielectric fluids are composed of nonpolar molecules, which lack positive or negative poles and exhibit low concentrations of free electrons and ions [80]. These chemical properties impede the flow of electrical current, giving dielectric fluids excellent electrical insulating capabilities [81]. Additionally, their chemical stability arises from stable covalent interactions between nonpolar molecules, making them resistant to chemical reactions with other substances [82]. These characteristics make dielectric fluids ideal for applications in electrical environments, such as electronics cooling, battery ESS, aerospace systems, semiconductor equipment, medical devices, and renewable energy systems, where they do not interfere with or react with surrounding electrical currents and mechanical components [83].

Compared to water, dielectric fluids exhibit several key differences. Their molecules are composed of hydrocarbon chains, resulting in weaker intermolecular forces compared to the hydrogen bonding in water [84]. This lack of polarity leads to lower surface tension, reducing

the energy barrier for vapor bubble formation and enabling bubbles to form more easily at lower superheats [85]. Weaker cohesive forces also promote more frequent bubble detachment at smaller diameters. However, dielectric fluids generally have much lower latent heat of vaporization than water, which limits their heat removal capability during boiling [28].

Dielectric fluids are primarily categorized into fluorinated fluids (e.g., fluorocarbons, hydrofluoroethers like Novec fluids) and non-fluorinated fluids (e.g., natural esters, silicone oils). Dielectric fluids, such as FC-72 [86], Novec 649 [87], Novec-7100 [88] and AE-3000 [89], are widely used in high performance applications involving electrical currents, owing to their excellent dielectric strength, chemical inertness, and low boiling points [90].

Table 2 presents a comparison of the thermophysical properties of various dielectric fluids (FC-72, Novec-7100, Opteon 2P50, ThermaSolv IM6) with those of water. Although dielectric fluids exhibit lower latent heat of vaporization, lower thermal conductivity, and reduced overall heat transfer performance compared to water, they remain attractive for many engineering applications. This is primarily because their electrical insulating capability, chemical stability, and compatibility with sensitive components outweigh the drawbacks in thermal performance. In systems where electrical safety and component reliability are paramount, such as electronics cooling, aerospace thermal management, and battery ESS, dielectric fluids offer unique advantages that cannot be matched by water or other conductive fluids.

As seen in Table 2, a drawback of conventional dielectric fluids such as FC-72 and Novec-7100, is that they possess high global warming potential (GWP), raising concerns regarding their contribution to climate change. Due to growing global concern over environmental sustainability, recently developed dielectric fluids have increasingly focused on minimizing GWP. The dielectric fluids introduced in Table 2, Opteon 2P50 [91] and ThermaSolv IM6 [92], were both recently specifically developed for the two phase cooling of electronics. Both fluids have boiling points between 47 and 49 °C, which aligns with the typical operating temperatures of electronics. They also have other similar thermophysical properties such as liquid density, viscosity, specific heat and latent heat of vaporization.

While both fluids have similar thermophysical properties, where Opteon 2P50 has slightly better heat transfer related properties compared to ThermaSolv IM6, there are differences in their environmental effects. However, ThermaSolv IM6 exhibits an ultra-low global warming potential GWP (≈ 1), whereas Opteon 2P50 has a low but still higher GWP (≈ 10) than ThermaSolv IM6. Notwithstanding, the GWP of both fluids are still significantly enhanced compared to conventional dielectric fluids such as FC-72 and Novec-7100. Regarding ozone depletion potential (ODP), both fluids have zero ODP as they do not possess chlorine or bromine atoms responsible for ozone destruction.

3.2. Influence of thermophysical properties on CHF and HTC

Based on the thermophysical properties listed in Table 2, the ratios of

Table 2

Comparison of Thermophysical Properties and Dimensionless Parameters of various fluids at Saturated Condition (1 atm).

	FC-72	Novec-7100	Opteon 2P50	ThermaSolv IM6	Water
Boiling point (°C)	57	61	48.9	47	100.2
Liquid density (kg/m^3)	1591.7	1510	1456	1600	957.3
Vapor density (kg/m^3)	12.08	-	-	-	0.59
Liquid dynamic viscosity ($\text{mPa}\cdot\text{s}$)	0.438	0.574	0.62	0.576	0.282
Liquid specific heat ($\text{J}/\text{kg}\cdot\text{K}$)	1103	1183	1090	1044	3798
Liquid thermal conductivity ($\text{W}/\text{m}\cdot\text{K}$)	0.0539	0.069	0.073	0.11	0.6751
Latent heat of vaporization (kJ/kg)	95.26	112	108	93	2267.91
Liquid surface tension (mN/m)	7.783	13.6	11	11.4	59.461
Liquid Thermal Expansion Coefficient ($1/\text{K}$)	0.00255	0.0018	-	-	0.000711
Global warming potential (GWP)	9300	320	10	1	-
Ozone Depletion Potential (ODP)	0	0	0	0	-

key dimensionless parameters—such as the Prandtl number (Pr), Bond number (Bo), Weber number (We), and Jakob number (Ja)—that influence global pool boiling heat transfer are summarized in Table 3. Here, parameters involving characteristic length and velocity scales are presented in ratio form to emphasize relative differences between fluids rather than absolute values, which can vary with the specific boiling configuration (see Table 4).

Main thermophysical properties listed in Table 2—including density ratio, vapor density, latent heat of vaporization (hfg), and surface tension—are plotted as functions of the reduced pressure (Pred) in Fig. 4. In Fig. 5, several dimensionless groups that characterize boiling and CHF behavior, such as the Weber, Bond, and Prandtl numbers, together with the Zuber-type CHF prediction listed in Table 3, are also displayed with respect to reduced pressure.

Considering the Weber and Bond numbers, FC-72 exhibits much higher values compared to water, which suggests that generated bubbles may be more vulnerable to external perturbations. Inertial and buoyancy forces could more readily overcome surface tension, potentially leading to more frequent and easier bubble departures. However, because its latent heat of vaporization and density difference are much smaller than those of water, each bubble is expected to remove less energy. As a result, although bubble activity may be more pronounced, both the theoretical CHF and HTC are likely to be significantly lower than that of water. Fig. 6 schematically illustrates these contrasting bubble departure characteristics in water and FC-72, highlighting the role of latent heat and surface tension in governing heat removal per bubble, bubble frequency, and the resulting differences in HTC and CHF.

4. CHF and HTC correlations

A variety of empirical and semi-empirical correlations have been proposed to predict the CHF and HTC in pool boiling. While many advanced models have been proposed for various conditions, the foundational models summarized below remain the basis for most predictions and are discussed in previous literatures [29,31,98,99].

4.1. Representative CHF correlations

4.1.1. Classical correlations

The mechanistic approaches for predicting pool boiling CHF have been extensively investigated, and their derivations and origins are well classified in previous literature (see Ref. [29] for a detailed summary). According to this reference, the major representative models include [1]: the bubble interference model, in which CHF is assumed to occur when growing vapor bubbles coalesce to cover the heated surface [2]; the hydrodynamic instability model, which attributes CHF to the merging of vapor columns driven by interfacial instabilities [3]; the macrolayer dryout model, where CHF is triggered by the complete evaporation of a liquid macrolayer beneath large coalesced bubbles; and [4] the interfacial lift-off model, which considers CHF to occur when vapor momentum lifts the bulk liquid away from the heated surface. Building on these legacy frameworks, subsequent studies have sought to improve predictive accuracy by incorporating surface-specific effects

Table 3
Comparison of Dimensionless Parameters of water and FC-72 Influencing Pool Boiling Heat Transfer.

	FC-72	Water	FC-72/ Water
Weber number ($We = \rho_f U^2 L / \sigma$)	-	-	12.7
Bond number ($Bo = g(\rho_f - \rho_g) L^2 / \sigma$)	-	-	12.6
Prandtl number ($Pr = \nu / \alpha$)	8.96	1.59	5.65
Jakob number ($Ja = \rho_f c_{p,f} \Delta T_{sub} / \rho_g h_{fg}$) [93]	-	-	0.562
Zuber CHF ($q_{CHF}^* = C_z \rho_g h_{fg} [\sigma g (\rho_f - \rho_g) / \rho_g^2]^{1/4}$)	170.9	1318.7	0.130

Table 4

Summary of CHF correlations accounting for subcooling and their experimental validation ranges.

Researchers	Correlations	Ranges confirmed by experiment	Test liquid
Kutateladze and Shneiderman [94]	$q_{CHF}^* / q_{sat}^* = 1 + 0.065(\rho_L / \rho_V)^{-1/5} Ja$	$0.1 \leq P \leq 1 \text{ MPa}$ $\Delta T_{sub} \leq 120 \text{ K}$	Water, Ethanol
Ivey and Morris [95]	$1 + 0.102(\rho_L / \rho_V)^{-1/4} Ja$	$P = 0.1 \text{ MPa}$ $\Delta T_{sub} \leq 70 \text{ K}$	Iso-octane
Zuber et al. [96]	$1 + 5.32 Pe^{-1/2} Ja$	$0.1 \leq P \leq 1 \text{ MPa}$ $\Delta T_{sub} \leq 120 \text{ K}$	Water, Ethanol
Mudawar and Anderson [93]	$1 + 0.0643(\rho_L / \rho_V)^{-1/4} Ja$	$0.1 \text{ MPa} \leq P \leq 0.3 \text{ MPa}$ $\Delta T_{sub} \leq 35 \text{ K}$	FC-72
Hwang and Moran (1981)	$1 + 0.05(\rho_L / \rho_V)^{-1/4} Ja$	—	FC-72

such as contact angle, wettability variations with pressure, and micro-scale surface structures [29].

- (i) **Bubble Interference (BI) Model** – First proposed by Rohsenow and Griffith [100], this model assumes CHF occurs when growing spherical bubbles interfere and merge, forming a continuous vapor layer. Rohsenow derived a semi-empirical correlation using dimensional analysis:

$$q_{CHF,R}^* = 0.012 \rho_g h_{fg} \left(\frac{\rho_f - \rho_g}{\rho_g} \right)^{0.6} \quad (1)$$

Although conceptually simple, high-speed photography later revealed that bubble coalescence precedes CHF, reducing predictive accuracy.

- (ii) **Hydrodynamic Instability (HI) Model** – Most notably represented by the Zuber correlation [101] attributes the onset of CHF to interfacial (Helmholtz) instabilities between adjacent vapor columns. As these instabilities grow, the columns coalesce, blocking the supply of liquid to the heated surface and leading to a vapor blanket formation. This model shares a similar form to the Kutateladze-type nondimensional correlation derived via dimensional analysis:

$$q_{CHF,Z-K}^* = K \rho_g h_{fg} \left[\frac{\sigma g (\rho_f - \rho_g)}{\rho_g^2} \right]^{1/4} \quad (2)$$

Zuber proposed $K = 0.131$, and several subsequent studies have adopted similar formulations with empirically fitted coefficient typically ranging from 0.13 to 0.17 [29,102–104]. This model has become the foundation of many models and numerous extensions incorporate effects of wettability, roughness, wall properties, pressure, viscosity, and orientation [29,31].

- (iii) **Macrolayer Dryout (MD) Model** – Proposed by Haramura and Katto [105], this mechanism links CHF to complete evaporation of a liquid macrolayer beneath large coalesced bubbles:

$$q_{CHF,H-K}^* = \rho_g h_{fg} \delta \left(1 - \frac{A_g}{A_w} \right) f \quad (3)$$

where δ is the initial thickness of the liquid macrolayer, f is the departure frequency of the bubbles, and the term A_g/A_w denotes the ratio of the cross-sectional area of the vapor stems to the total heater surface area. Consequently, the term A_g/A_w corresponds to the fraction of the heater surface that is initially wetted by the liquid macrolayer.

This approach inspired studies on macrolayer thickness estimation, although some experiments question whether stable macrolayers exist at CHF [29].

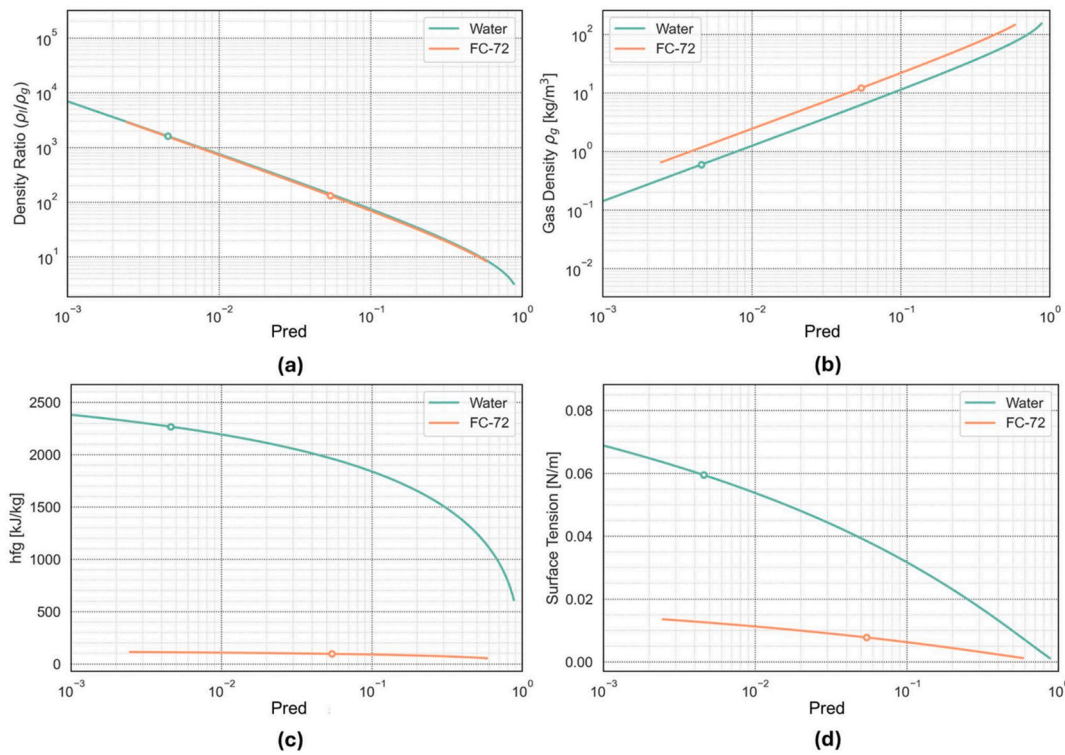


Fig. 4. Variation of key thermophysical properties of water and FC-72 (density ratio, vapor density, latent heat of vaporization, and surface tension) with reduced pressure (Pred).

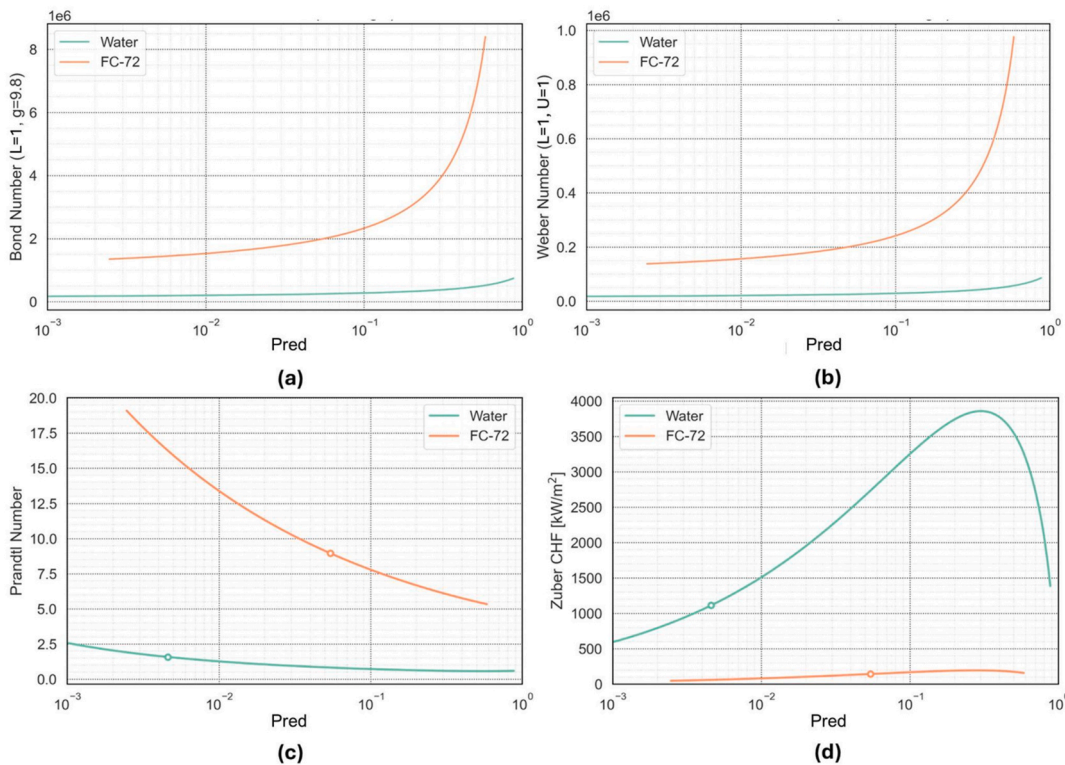


Fig. 5. Dimensionless groups relevant to boiling and CHF of water and FC-72 (Weber, Bond, and Prandtl numbers) and Zuber-type CHF prediction plotted against reduced pressure (Pred).

(iv) **Interfacial Lift-off (IL) Model** – Developed by Galloway and Mudawar [106,107], CHF occurs when vapor momentum is sufficient to lift the bulk liquid away from the heated surface. Originally

proposed for flow boiling, this mechanism was later observed in pool boiling as well, where strong vapor jets disrupt liquid contact [108]. Guan et al. [68]. adapted this concept to upward-facing surfaces:

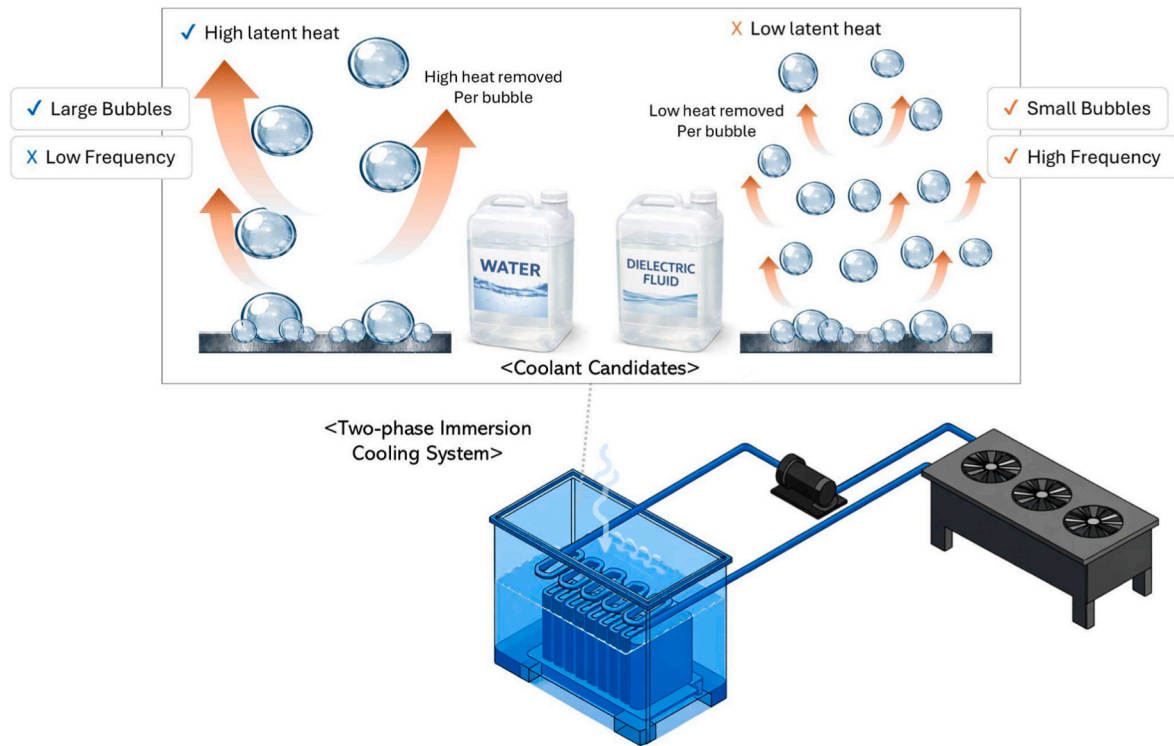


Fig. 6. Schematic comparison of bubble departure characteristics and heat removal efficiency governing HTC and CHF in water and FC-72. (The schematic of two-phase immersion cooling system was reproduced from previous literature [97].)

$$q_{CHF,G}'' = 0.2445 \left(1 + \frac{\rho_g}{\rho_f} \right)^{\frac{1}{4}} \left(\frac{\rho_g}{\rho_f} \right)^{\frac{1}{10}} q_{CHF,Z-K}'' \quad (4)$$

Under saturated conditions, this often reduces to the Kutateladze–Zuber form.

4.1.2. Extended correlations incorporating pressure and subcooling

While classical models such as those proposed by Zuber–Kutateladze perform well under saturated conditions and for large horizontal surfaces, they often fail to capture the effects of system pressure and subcooling across a wide range of fluids. To overcome these limitations, many researchers have proposed modified correlations that explicitly incorporate pressure and subcooling influences.

(i) Effect of Pressure

Earlier efforts have established that classical correlations can accurately predict CHF for helium and other cryogenic fluids when the coefficients are appropriately adjusted. Deev et al. [109] introduced a modified Zuber–Kutateladze-type correlation for helium that retained the original dimensional form but required a higher coefficient (e.g., $K = 0.2$) to fit experimental data at reduced pressures ($Pred < 0.75$). This adjustment indicates that while the general CHF scaling with thermophysical properties remains valid, pressure-specific calibration is essential for accurate prediction. Bewilogua et al. [103] extended this approach by compiling CHF data over a wide range of reduced pressures for various cryogenic fluids, including liquid helium and hydrogen. Their results demonstrated good agreement with the Zuber–Kutateladze correlation and revealed that CHF typically reaches a maximum at a reduced pressure of $Pred \approx 0.35$. They further proposed a polynomial form to represent the CHF dependence on pressure. This trend was later confirmed in the recent study by Wang et al. [110] who also observed a CHF peak near $Pred \approx 0.35$ for liquid hydrogen and suggested a polynomial correlation specific to high-pressure cryogenic boiling behavior. Moreover, the CHF dependence on pressure was suggested by the

correlation:

$$q_{CHF}'' = \left[0.18 - 0.14 \left(\frac{P}{P_c} \right)^{5.68} \right] \rho_g h_{fg} \left[\frac{\sigma g (\rho_f - \rho_g)}{\rho_g^2} \right]^{1/4} \quad (5)$$

In contrast, for low-wetting organic fluids such as ethanol and R-141b, the Zuber correlation has been shown to maintain reasonable accuracy across a wide pressure range. Bailey et al. [111] investigated the effect of system pressure on CHF for pentane, methanol, and water at pressures up to 7 atm. While their results also showed good agreement with the Zuber–Kutateladze model, a slightly higher coefficient was required. However, it is worth noting that their experimental pressure range was limited to relatively low pressures. Sakashita [112] observed that the wettability of fluids like ethanol and R-141b is relatively insensitive to pressure, allowing classical models such as Zuber–Kutateladze to predict CHF with minimal adjustments. In contrast, the wettability of water increases with pressure, leading to deviations from classical predictions. His CHF data for water aligned more closely with the correlation by Kirichenko and Chernyakov [113], which accounts for pressure dependent wettability effects. Such pressure and surface-sensitive extensions including those addressing wettability have been systematically reviewed by Liang et al. (2018) [29], who highlighted the need for targeted modifications to classical frameworks to improve generality across fluid types and pressure regimes.

(ii) Effect of Subcooling

The influence of subcooling on CHF has often been represented by a linear enhancement term. Ivey and Morris [95] proposed one of the most widely used forms:

$$q_{CHF}'' = q_{sat}'' (1 + C_{CHF,sub} \Delta T_{sub}) \quad (5)$$

Many subsequent studies adopted this formulation with different values of the subcooling coefficient. With respect to dimensionless parameters, researchers frequently employ the Jakob number (Ja) to

capture subcooling effects, as it represents the ratio of sensible to latent heat. For water, several early correlations were formulated in terms of Ja, occasionally supplemented by the Peclet number (Pe) to account for thermal diffusion effects at high subcooling levels. However, for dielectric fluids such as FC-72, which typically exhibit lower Prandtl numbers and weaker thermal diffusion near the wall, the role of Pe appears less pronounced.

Nevertheless, most extensions incorporating subcooling consistently demonstrate that CHF increases with greater liquid subcooling. This enhancement is attributed to delayed onset of the vapor blanket formation and improved liquid replenishment near the heated surface, both of which strengthen boiling stability.

4.2. Representative HTC correlations

The correlations for the prediction of nucleate pool boiling HTC can be developed through two types of methodologies. The first type is based on fluid properties and is refined by fitting empirical constants and exponents. The second type additionally includes fluid-surface effects. For research which studies surface modification to enhance boiling heat transfer, this method provides a more accurate representation of practical boiling situations and therefore achieves higher predictive reliability. This makes it more widely adopted in recent research [114]. The following correlations represent some of the earliest and most fundamental models for predicting HTC and form the foundation for many subsequent developments.

Kruzhilin (1947) [115] developed the earliest known HTC correlation, by using the first methodology. The equation was formulated through the dimensionless analysis of nucleate boiling, where the HTC was scaled with the fluid thermophysical properties. This reflects the assumption that bubble dynamics is the primary factor in determining HTC. The resulting correlation is expressed as:

$$h = 0.082 \left(\frac{k_f}{L_b} \right) \left(\frac{T_{sat} c_{p,f} \sigma \rho_f}{h_{fg}^2 \rho_g^2 L_b} \right)^{0.33} Pr_f^{-0.45} \left(\frac{h_{fg} \rho_f q''}{g T_{sat} h_{fg}} \right)^{0.7} \quad (6)$$

where $L_b = \sqrt{\sigma / (g(\rho_f - \rho_g))}$, Pr_f is the Prandtl number. The numerical constant 0.082 was derived from the regression of experimental data of saturated nucleate pool boiling of water at various pressures and on plain surfaces. Therefore, the effects of surface roughness and fluid-surface combinations are not reflected, arising the need for revised correlations.

Rohsenow (1952) [100] proposed one of the most widely used HTC correlations, by using the second methodology. Similar to the Kruzhilin correlation, it assumes that the motion of bubbles at the instant of departure from the heated surface is the primary factor in determining HTC. It also adopts the bubble agitation mechanism, in which the separation and movement of bubbles from the surface lead to liquid counterflow towards the surface, enhancing heat transfer. The resulting correlation is expressed as:

$$h = \left(\frac{\mu_f}{L_b} \right)^r \frac{c_{p,f}}{C_{s,f}} Pr_f^{-s} \left(\frac{q''}{h_{fg}} \right)^{1-r} \quad (7)$$

where $L_b = \sqrt{\sigma / (g(\rho_f - \rho_g))}$, $C_{s,f}$ is an empirical constant depending on the surface–fluid combinations, Pr_f is the Prandtl number, $r = \frac{1}{3}$, $s = 1$ for water and $s = 1.7$ for other fluids. Unlike the Kruzhilin correlation, the effects of surface roughness and fluid-surface combinations are reflected through the empirical constant $C_{s,f}$, allowing the Rohsenow correlation to provide more practical results and therefore making it still widely used up to today.

McNelly (1953) [116] proposed the HTC correlation based on the idea that the thin liquid film on the heated surface concentrated resistance. Within this film, the rapid generation of vapor bubbles creates

turbulence, which was assumed to have major effects on heat transfer. This assumption, together with dimensional reasoning, was used to formulate a correlation that highlights bubble induced agitation, particularly the early growth phase, as the primary mechanism for enhancing heat transfer. The resulting correlation is expressed as:

$$h = 0.225 \left(\frac{\mu_f k_f}{\sigma} \right)^{0.31} \left(\frac{\rho_f}{\rho_g} - 1 \right)^{0.31} Pr_f^{-0.45} \left(\frac{c_{p,f} q''}{h_{fg}} \right)^{0.69} \quad (8)$$

where Pr_f is the Prandtl number. Rather than just dimensionless property groups, bubble scale mechanisms are also reflected in the correlation, making it more related to the mechanisms of nucleate boiling.

Förster and Zuber (1955) [117] developed the HTC correlation by utilizing bubble radius and growth dynamics near the heated surface. It was indicated that while smaller bubbles grow rapidly and larger ones slowly, the degree of agitation in the surrounding liquid due to bubble growth remains the same. Also, the product of the bubble's growth rate and its radius remain nearly constant. This insight led to the definition of a Reynolds number for liquid motion adjacent to the surface. In the correlation, the Nusselt number is expressed as a function of Reynolds and Prandtl numbers. The resulting correlation is expressed as:

$$h = \frac{0.00122 \Delta T_{sat}^{0.234} \Delta p_{sat}^{0.75} c_{p,f}^{0.45} \rho_f^{0.49} k_f^{0.79}}{\sigma^{0.5} h_{fg}^{0.24} \mu_f^{0.24} \rho_f^{0.29} \rho_g^{0.24}} \quad (9)$$

where Δp_{sat} represents the difference between the saturation pressure at the wall temperature and that at the bulk liquid temperature. Unlike earlier approaches that considered only bulk fluid properties, this also reflects the influence of pressure in the correlation.

Kutateladze (1959) [118] developed the HTC correlation based on the assumption that balance between surface tension and buoyancy defines the bubble size, which plays a significant role in determining HTC. In addition, it was also assumed that heat transfer from the surface to the bubble occurs primarily through the liquid microlayer between the surface and the bubble. The dimensional analysis based on these assumptions produced nondimensional groups consisting of factors including the Prandtl number, density ratio, and Laplace length. The resulting correlation is expressed as:

$$h = 7 \times 10^{-4} \left(\frac{k_f}{L_b} \right) Pr_f^{0.35} \left(\frac{P L_b}{\sigma} \right)^{0.7} \left(\frac{\rho_f}{\rho_g} \right)^{0.7} \left(\frac{q'' L_b}{\mu_f h_{fg}} \right)^{0.7} \quad (10)$$

where $L_b = \sqrt{\frac{\sigma}{g(\rho_f - \rho_g)}}$, P is the system pressure, Pr_f is the Prandtl number. Similar to the Kruzhilin correlation, the numerical constant 7×10^{-4} was obtained from water boiling experimental data of plain surfaces. Similar to the Förster and Zuber correlation, the influence of pressure is considered.

Cooper (1984) [119] introduced the HTC correlation that relates boiling heat transfer to reduced pressure, heat flux, fluid molecular weight, and surface roughness. To simplify the application of fluid thermophysical properties, they are expressed through reduced pressure and reduced temperature, the latter evaluated from the former using the Clapeyron relation. The molecular weight of the fluid is incorporated to capture the effects of viscosity and thermal conductivity, with approximate proportionalities. The surface roughness is represented as a logarithmic exponent of reduced pressure, as it has a larger influence at low pressures.

$$h = 55 \times M^{-0.5} P_R^m (-\log(P_R))^{-0.55} q''^{0.67} \quad (11)$$

where $m = 0.12 - 0.2 \times \log(R_p)$, P_R is the reduced pressure, R_p is the surface roughness (Cooper recommends $R_p = 1 \mu m$ if unknown), M is the molecular weight of the liquid. The numerical constant 55 is derived from the regression of the experimental database of various fluids, pressures and surface roughness.

5. Pressure and subcooling effects on CHF

This section addresses how pressure and subcooling conditions govern the CHF, with the subsequent section focusing on HTC. Most investigations of these effects have been conducted with water, while only a limited number of systematic studies exist for representative dielectric fluids such as FC-72, HFE-7100, and AE-3000, as summarized in Table 5.

Consistent with the scope of this study, which is strictly restricted to plain surfaces to establish a fundamental baseline, all data points in these figures represent non-modified specimens. The performance variations illustrated in the plots of Fig. 7 demonstrate that even without surface modification, the thermal limits of dielectric fluids can be significantly extended through the synergy of operating parameters and heater material properties. Specifically, the highest thermal limits reported in the literature—such as those by O'Connor et al. [44] and Dedov et al. [131]—are attained when high system pressure or

significant liquid subcooling (e.g., up to 85.22 °C) is employed on high-thermal-conductivity materials like copper.

Pressure primarily influences bubble nucleation, growth, and detachment, as well as liquid replenishment and surface wettability, thereby shaping CHF behavior across different fluids and operating regimes. Subcooling, by contrast, enhances liquid supply to the heated surface and delays the onset of vapor blanket formation, generally leading to higher CHF values. By analyzing the combined roles of pressure and subcooling, a more comprehensive understanding of CHF mechanisms can be achieved. The following subsections first compare pressure-dependent CHF trends for water and dielectric fluids and then examine how liquid subcooling modifies these trends.

5.1. Pressure dependence of CHF: water vs. dielectric fluids

5.1.1. Pressure dependence of CHF on water

Water has been the primary working fluid in CHF research, and its

Table 5

Summary of CHF and HTC measurements for water and dielectric fluids under varying pressure and subcooling.

Reference	Specimen type (plain)	Heater type	Specimen heater size (mm × mm)	Fluid type	Subcooling ΔT (°C)	System pressure (kPa)	CHF (kW/m ²)	Max. HTC (kW/m ² ·K)
Kim et al. (2002) [120]	Quartz	Deposited platinum layer	2.7 × 2.7	FC-72	8.6-33.9	101.33	179.1-281.3	6.53-7.33
Raj et al. (2012) [121]	Quartz (SiO ₂ coated)	Deposited platinum layer	7 × 7	FC-72	1-26	58.77-188.46	101-226.6	2.89-7.82
Wang et al. (2022) [122]	Silicon	Soldered 2 copper wires	10 × 10, 20 × 20	FC-72	41	101.33	205.7, 289.5	2.7, 3.7
Raj et al. (2010) [123]	Quartz	Deposited platinum layer	7 × 7	FC-72	6.6, 16.6	101.33	176.2, 208.8	6.8, 8.2
Di Marco et al. (2009) [124]	Zinc Sulfide	Sputtered gold layer	20 × 20	FC-72	12.6	101.33	159.4	4.1
Zhao et al. (2009) [125]	Al ₂ O ₃ ceramic	Multi-layer alloy (Cr, Cu, Ni, Au)	15 × 15	FC-72	33	101.33	246	7.68
Raj et al. (2010) [126]	Quartz	Deposited platinum layer	7 × 7	FC-72	8.6, 26.6	101.33	228.8, 232.9	5.66, 5.82
Rainey et al. (2003) [77]	Copper	Sputtered tantalum and titanium nitride layers	10 × 10	FC-72	0	30-150	115-173	4.87-10.79
Garivalis et al. (2024) [127]	Silicon	P-doped N-type silicon	10 × 10	FC-72	5	101.33	178.8	5.26
Dahariya et al. (2019) [66]	Copper	Cartridge heater	11 (diameter)	Water	0	101.33-515.03	1147.2-1505.7	47.6-117.3
Jun et al. (2018) [75]	Copper	Soldered resistive heater	10 × 10	Water	0-30	101.33	1070-2820	33.94-70.2
Parker et al. (2005) [128]	Copper	Heating element attached with epoxy	10 × 10	FC-72	0-30	101.33	169-295	8.83-14.06
Wei et al. (2003) [129]	P doped N-type silicon	Soldered 2 copper wires	10 × 10	FC-72	0-45	101.33	162.37-294.78	4.62-6.86
O'Connor et al. (1996) [44]	Copper (polished)	Sputtered platinum layer	5 × 16	FC-72	0-85.22	101.33, 303.99	158-480	5.10-15.19
El-Genk et al. (2004) [130]	Copper	Wound nichrome wire	10 × 10	Novec-7100	0-30	101.33	200-314.82	9.95-12.26
Dedov et al. (2025) [131]	Copper	Cartridge heater	30 (diameter)	Novec-7100	0	303.99, 405.32	307.61, 342.05	16.23, 20.48
Zhou et al. (2019) [132]	P-doped N-type silicon	Soldered 2 copper wires	10 × 10	FC-72	0-25	101.33	155.55-221.47	4.79-6.57
Sathyamurthi et al. (2009) [133]	Silicon	Cartridge heaters	59 × 32	PF-5060	0-30	101.33	93.82-134.24	5.69-6.38
Wei et al. (2005) [134]	Silicon	Soldered 2 copper wires	10 × 10	FC-72	25, 45	101.33	220.7, 291.6	5.77, 7.23
Ono et al. (2007) [135]	Copper (polished)	Cartridge heaters	8 (diameter)	Water	0-30	101.33	2080.48-3854.72	75.35-110.60
Fan et al. (2022) [136]	Copper (polished)	Cartridge heaters	40 (diameter)	Novec-7100	0	70-200	148.66-242.20	5.28-9.18
Yu et al. (2024) [42]	Copper (polished)	Ceramic heater	20 × 20	Novec-7100	0	8.8-97.1	128.03-210.74	3.28-7.03
Gao et al. (2023) [137]	Copper	Cartridge heater	11 (diameter)	Water	0-20	101.33	1002.22-52.65	52.65-77.65
Bailey et al. (2006) [111]	Copper (nickel coated, polished)	Cartridge heater	10 × 10	Water	0	20-300	531.98-2605.04	18.95-96.89
Wu et al. (2025) [138]	Sapphire (TiO ₂ coated)	ITO film	10 × 10	AE-3000	0	101.33	101.61	3.19
Kano (2015) [139]	Copper (nickel and boron coated)	Cartridge heaters	15 (diameter)	AE-3000	0	101.33	214.94	8.53

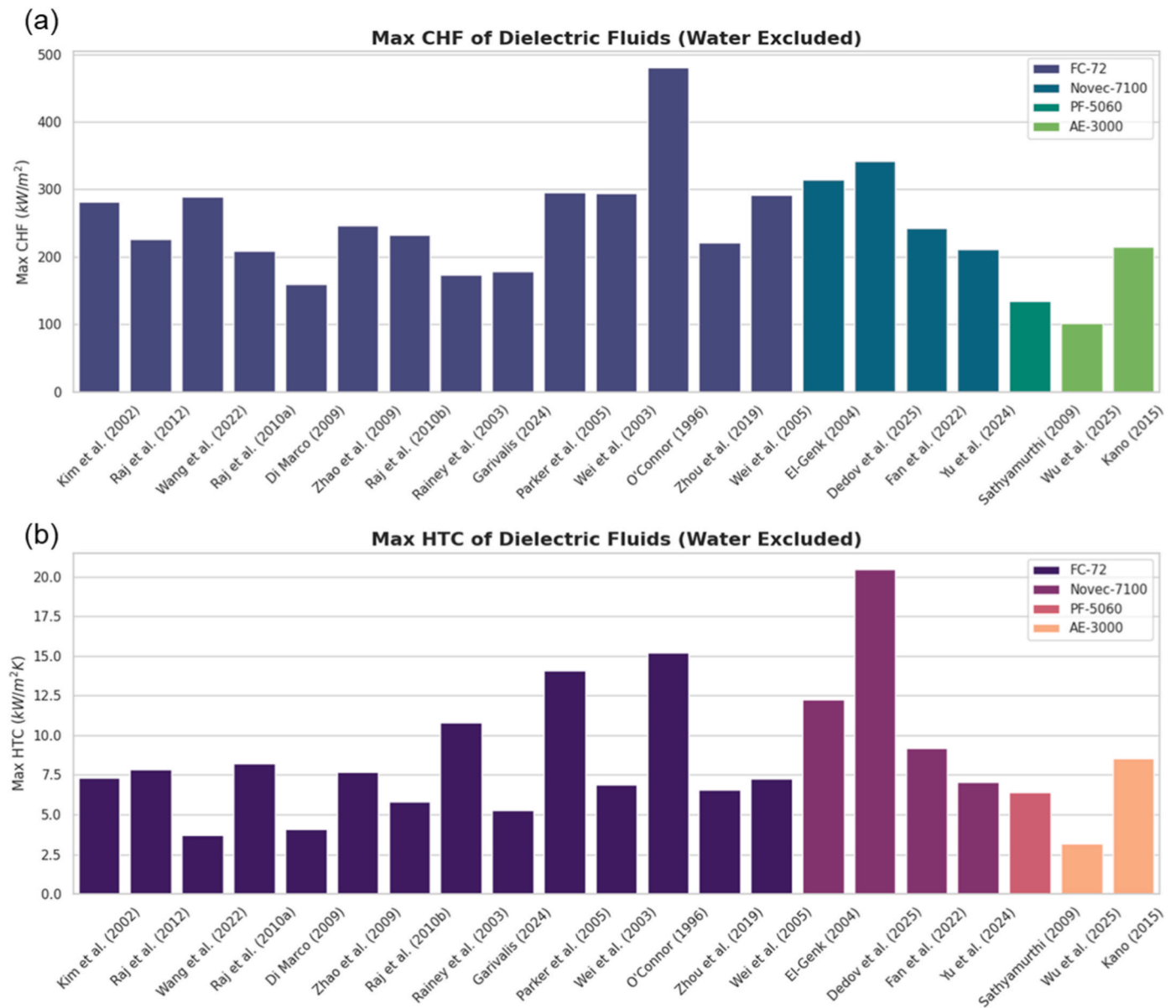


Fig. 7. Comparison of maximum thermal performance for various dielectric fluids on plain surfaces under different operating conditions: (a) CHF and (b) HTC.

behavior across a wide pressure range has been extensively documented. Early studies established the Kutateladze–Zuber hydrodynamic-instability framework as the classical basis, but subsequent investigations revealed systematic deviations at elevated pressures.

Sakashita and Ono [64] demonstrated that CHF on horizontal copper plates increases with pressure from 0.1 to 7 MPa, with macrolayer dryout remaining a valid mechanism even in high-pressure regimes. Sakashita [112] further highlighted that CHF on vertical surfaces grows more steeply with pressure than predicted by classical correlations, emphasizing that wettability variations with pressure are crucial. Dahariya & Betz [66] and Dahariya et al. [69] extended these observations: they reported significant CHF enhancement under elevated pressures (up to ~0.4 MPa for smooth plates and ~0.2 MPa for sintered wick surfaces), attributing the increase to shorter instability wavelengths, bubble pinning, and capillary assisted liquid replenishment. Complementary transient measurements by Walunj & Sathyabhama [140] confirmed that CHF rises with pressure (1–10 bar) under impulsive heating, although the magnitude decreases for longer pulses. Bailey et al. [111] also found a power-law dependence of CHF on saturation pressure for polished copper plates, consistent with Zuber-type scaling

but requiring a modified constant ($K \approx 1.7$).

These findings converge on a common limitation: legacy correlations (Rohsenow, Zuber, Bailey) do not provide universal accuracy across different pressures. Two strategies are suggested as a solution. The first involves wettability based corrections to Zuber-type formulations, as emphasized in Sakashita [112] and Dahariya [66,69]. Sakashita [112] showed that CHF data for water align more closely with the correlation of Kirichenko and Chernyakov [113], which explicitly incorporates pressure-dependent wettability effects. Consistently, Dahariya et al. [66] highlighted the critical role of wettability and recommended the use of models such as Kandlikar [141], which directly account for wettability variations. While these modifications significantly improve predictive accuracy, their applicability remains case-specific due to strong surface dependence.

The second strategy was to apply alternative physically based models, such as Guan's interfacial lift-off model, Rajvanshi's macrolayer based model, and Kumada's microlayer informed model, each showing better alignment with high pressure CHF data.

In this study, six representative correlations are compared: Rohsenow (bubble interference), Zuber and Bailey (hydrodynamic instability),

Guan (lift-off), Rajvanshi (macrolayer), and Kumada (microlayer).

Fig. 8 compiles water CHF data against the reduced pressure (Pred) and compares them with six representative correlations—Rohsenow (R), Zuber (Z), Bailey (B), Kumada (K), Rajvanshi (Ra), and Guan (G)—covering four modeling frameworks. Filled symbols denote hot-plate measurements, whereas half-filled symbols correspond to wire data; the two geometries display distinct characteristics. In Fig. 8, it can be seen that measurements approaching the critical pressure remain sparse.

For hot-plate data, the Zuber correlation matches reasonably well at low pressures but significantly underpredicts CHF at elevated pressures, consistent with the previous studies. This underprediction is generally attributed to the strong sensitivity of water wettability to pressure, which the hydrodynamic-instability framework does not capture. In contrast, wire data sometimes align better with Zuber despite its plate-based derivation, likely reflecting the reduced influence of wettability on wire geometries, where bubble departure and liquid replenishment are less sensitive to contact angle. However, extremely fine wires may exhibit early dryout due to their limited thermal inertia, yielding CHF values below Zuber's prediction. The Bailey correlation, also within the hydrodynamic instability family, provides slightly higher predictions than Zuber but still underestimates CHF at high pressures. Similar underprediction is observed for the Rohsenow (bubble interference model) and Guan (interfacial lift-off model) correlations. To improve predictive accuracy, Rajvanshi and Kumada proposed modified formulations that show improved agreement in the high-pressure regime, though Rajvanshi tends to overpredict CHF at low pressures and Kumada shows slight underprediction at elevated pressures. Wettability corrected variants of the hydrodynamic instability models have been proposed and shown better agreement, but their case specific nature limits general applicability. Accordingly, this study adopts the more broadly applicable Rajvanshi and Kumada correlations for comparative evaluation.

As noted by Sakashita and other researchers, heater geometry exerts a strong influence on CHF. Wire thickness is particularly critical: very fine wires often yield CHF below Zuber's prediction due to early dryout, while intermediate-thickness wires and ribbons show closer agreement. In contrast, thicker wires and plate-like geometries systematically diverge from Zuber at high pressures, reflecting wettability effects absent from the hydrodynamic-instability framework. By compiling and contrasting these datasets, Figs. 9 and 10 make these geometry-driven tendencies much more evident. The corresponding quantitative error metrics for plate-type and non-plate heaters are summarized in Tables 6 and 7, respectively, where mean absolute error (MAE) and mean

absolute percentage error (MAPE) of the datasets are defined as:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - x_i|, MAPE = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - x_i}{x_i} \right| \quad (12)$$

where x_i and y_i denote the measured and predicted values, respectively, and N is the total number of data points.

5.1.2. Pressure dependence of CHF on dielectric fluids

In contrast to water, dielectric fluids such as FC-72 and HFE-7100 possess intrinsically lower surface tension, smaller latent heat of vaporization, and reduced liquid–vapor density differences, which render them highly sensitive to pressure variations. Nevertheless, available experimental data remain scarce, especially at elevated pressures, limiting systematic comparisons.

For FC-72, several studies have addressed the pressure dependence of CHF. Arik and Bar-Cohen [142] investigated FC-72 and FC-40 binary mixtures and reported that CHF consistently increased as system pressure rose from 1 to 3 bar. Rainey et al. [67] examined FC-72 over the range of 30–150 kPa and confirmed that CHF increases with pressure on both plain and microporous surfaces. Guan et al. [68] measured saturated pool-boiling CHF of FC-72 at 150, 300, and 450 kPa, finding a monotonic increase with pressure that was well predicted by their interfacial lift-off model. Zhao et al. [125] further demonstrated in microgravity conditions that CHF of FC-72 increases with both system pressure (40–100 kPa) and subcooling, though the two effects could not be completely separated. For HFE-7100, experimental studies similarly highlight a strong sensitivity of CHF to pressure as well as surface conditions. Fariñas Alvarino et al. [143] reported that CHF increases with system pressure between 25 and 200 kPa, with the enhancement becoming more pronounced on rougher surfaces. Fan et al. [136] further confirmed a clear CHF increase as pressure rose from 0.7 to 2.0 bar, noting that the pressure effect was more dominant than surface roughness.

With the datasets compiled to date as seen in Fig. 11, predictive trends for dielectric fluids deviate systematically from those established for water. Among the evaluated correlations, the Bailey model exhibits the closest overall agreement with measured CHF across the investigated reduced-pressure range, followed by the Guan model. As demonstrated in Figs. 12 and 13, Bailey's predictions for various heater types and surface conditions generally fall within $\pm 20\%$ of the experimental data. In contrast, the classical Zuber correlation consistently underpredicts CHF, whereas the Rohsenow correlation tends to overpredict.

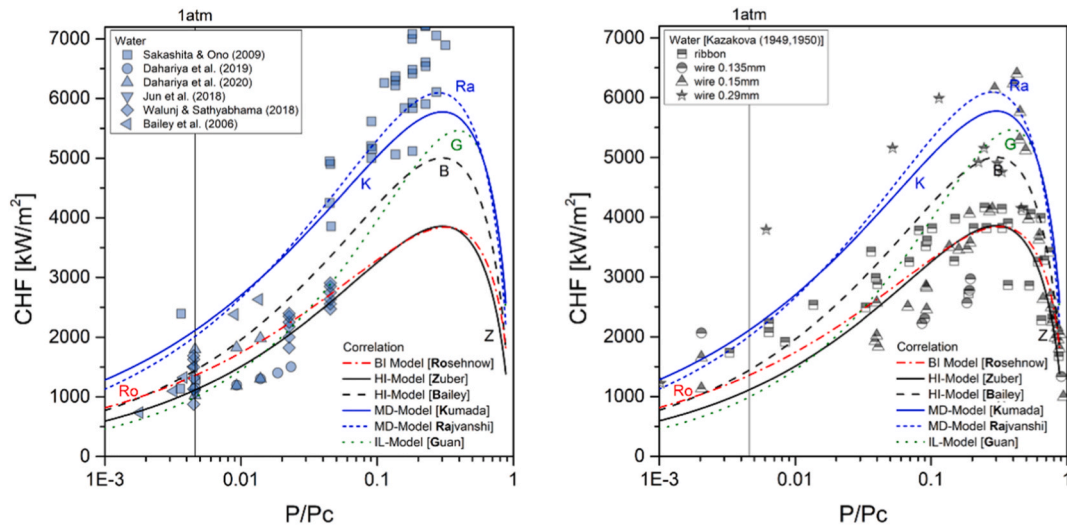


Fig. 8. CHF of water as a function of reduced pressure (Pred): comparison with six representative correlations. Left: plate-type heaters; Right: non-plate geometries (wires and ribbons).

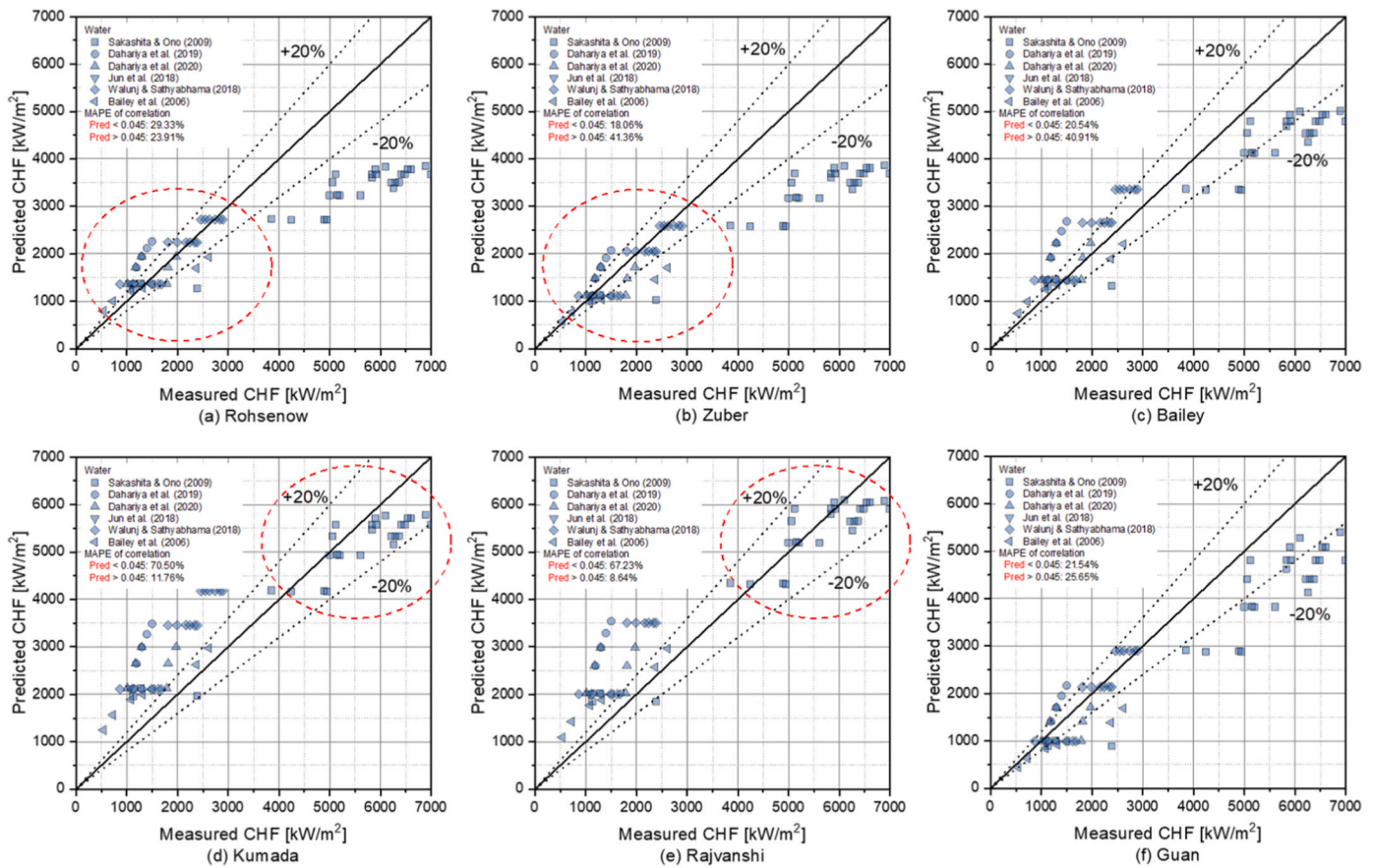


Fig. 9. Predicted versus measured CHF for water on plate-type heaters: comparison across six representative correlations.

Among the correlations evaluated, the Bailey model demonstrated the best agreement with experimental CHF data for both FC-72 and HFE-7100, a trend that is particularly evident in Figs. 12 and 13. Although still based on empirical extensions of hydrodynamic theory, Bailey's formulation accounts more effectively for fluid-property differences and provides closer alignment with the observed pressure dependence. While the Guan model proved better agreement for high-pressure water cases, its predictive accuracy for FC-72 was limited. These tendencies are quantitatively confirmed by the error metrics summarized in Tables 8 and 9, where the Bailey model exhibits the lowest mean absolute MAE and MAPE for both FC-72 and HFE-7100.

Overall, the comparative analysis between water and dielectric fluids underscores that the role of pressure in CHF behavior is fluid dependent and cannot be generalized solely based on water data. While water exhibits a clear shift in model applicability from low to high pressures, dielectric fluids reveal a systematic underestimation by classical models and require correlations that account for their unique thermophysical properties. The limited availability of experimental results for dielectric fluids such as FC-72 and HFE-7100 remains a critical barrier, and future studies should prioritize expanding pressure dependent CHF database for these fluids. Such efforts will not only improve the fidelity of existing models but also enable the formulation of universal frameworks capable of capturing CHF trends across a wide spectrum of working fluids and operating conditions.

5.2. Subcooling dependence of CHF: water vs. dielectric fluids

5.2.1. Subcooling dependence of CHF on water

Comparative studies of subcooling effects in water and dielectric fluids are relatively scarce, and only a few reports highlight systematic differences. In fact, systematic CHF datasets for subcooled water are

more limited than expected given the extensive research on saturated pool boiling. This is largely because most early water studies were driven by power-generation applications dominated by saturated conditions.

More recent investigations, however, have highlighted how subcooling can significantly enhance CHF in water boiling. For example, Gao et al. [137] reported that on micro-pillar structured surfaces, increasing subcooling intensifies natural convection, suppresses bubble growth and coalescence, and thereby increases CHF. Similarly, Jun et al. [75] observed that CHF increases nearly linearly with subcooling on both HTCMC (high-thermal-conductivity metal-ceramic composite) and plain copper surfaces, with a proportion of approximately 60 kW/m² per degree of subcooling. These findings illustrate that, while the database for subcooled water CHF remains limited, subcooling can exert a clear and quantifiable influence, particularly on modified surfaces.

Available datasets shown in Fig. 14 confirm these trends. While CHF–subcooling comparisons are often normalized by the saturated heat flux q''_{sat} , many datasets do not provide explicit values. To ensure consistency, CHF values in this study were therefore normalized by the Zuber heat flux (q''_{zube}). For analysis, the Zuber correlation and Kutateladze correlation (originally developed for water/ethanol) and the Mudawar correlation (empirically fitted for FC-72) were applied. As shown in Fig. 14, a pronounced divergence emerges: correlations validated for FC-72, fail to capture water behavior.

5.2.2. Subcooling dependence of CHF on dielectric fluids

Several experimental investigations have examined the role of subcooling in FC-72 boiling; however, most of these studies were not only focused on subcooling, but also combined it with other influencing factors such as pressure variation or gravity level. As a result, while a consistent trend of CHF enhancement with increasing subcooling has

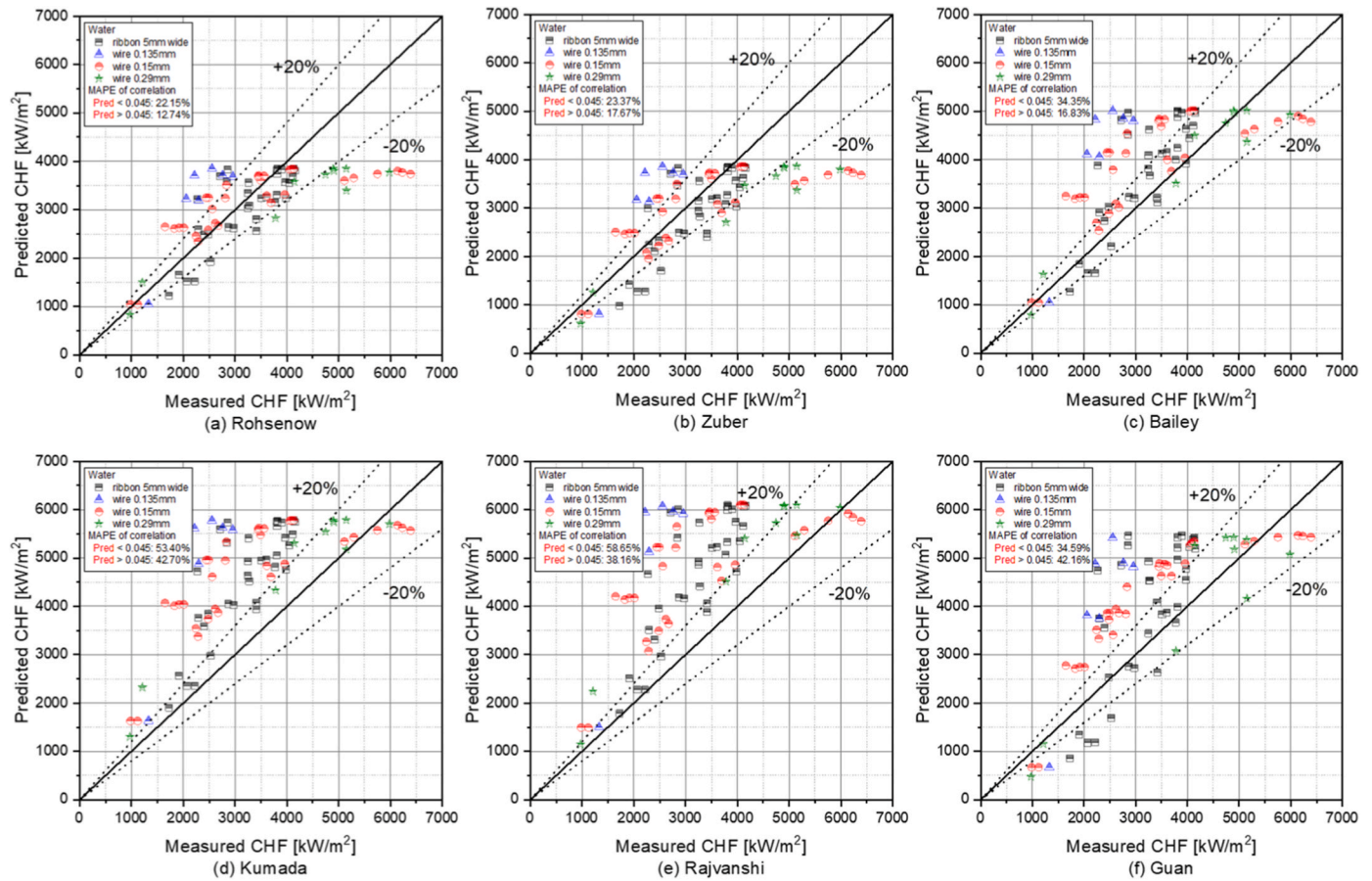


Fig. 10. Predicted versus measured CHF for water on non-plate heaters (wires and ribbons): comparison across six representative correlations.

Table 6

MAE and MAPE of representative CHF correlations for water on plate-type heaters, evaluated separately for low- and high-reduced-pressure regimes.

Low pressure regime (Pred < 0.045)						
	Rohsenow	Zuber	Bailey	Kumada	Rajvanshi	Guan
MAE [kW/m ²]	0.3013	0.3121	0.4658	1.0641	1.0471	0.3589
MAPE [%]	29.3289	18.0610	20.5394	70.4997	67.2348	21.5422
High pressure regime (Pred > 0.045)						
	Rohsenow	Zuber	Bailey	Kumada	Rajvanshi	Guan
MAE [kW/m ²]	2.4680	2.4868	1.4574	0.7371	0.5321	1.5275
MAPE [%]	23.9076	41.3641	40.9135	11.7622	8.6397	25.6504

Table 7

MAE and MAPE of representative CHF correlations for water on non-plate heaters, evaluated separately for low- and high-reduced-pressure regimes.

Low pressure regime (Pred < 0.045)						
	Rohsenow	Zuber	Bailey	Kumada	Rajvanshi	Guan
MAE [kW/m ²]	0.7352	0.7647	0.99215	1.4958	1.6731	0.9889
MAPE [%]	22.1508	23.3665	34.3525	53.4005	58.6505	34.5873
High pressure regime (Pred > 0.045)						
	Rohsenow	Zuber	Bailey	Kumada	Rajvanshi	Guan
MAE [kW/m ²]	0.4109	0.5446	0.4645	1.1715	1.0798	1.1598
MAPE [%]	12.7448	17.6735	16.8292	42.7034	38.1620	42.1634

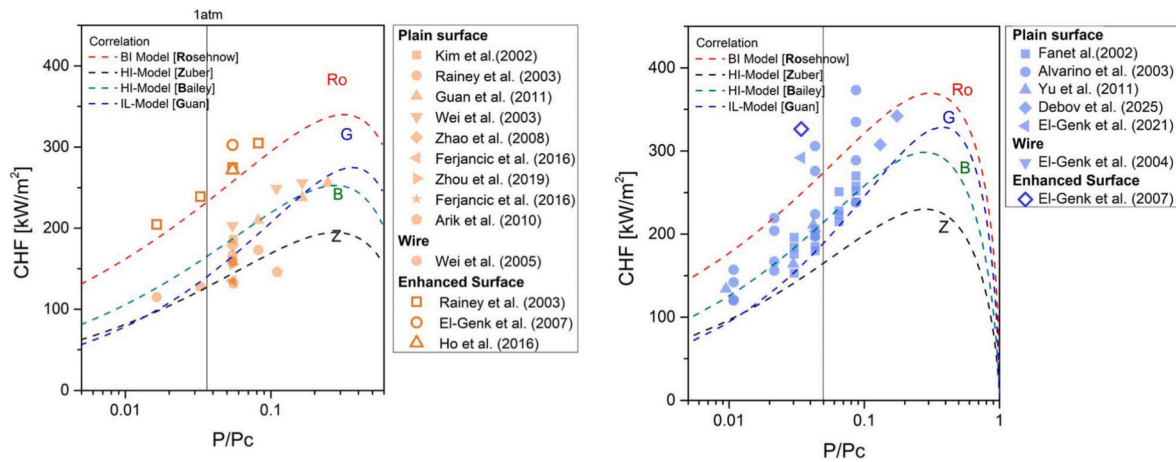


Fig. 11. CHF of FC-72 (Left) and HFE-7100 (Right) as a function of reduced pressure (Pred): comparison with representative correlations.

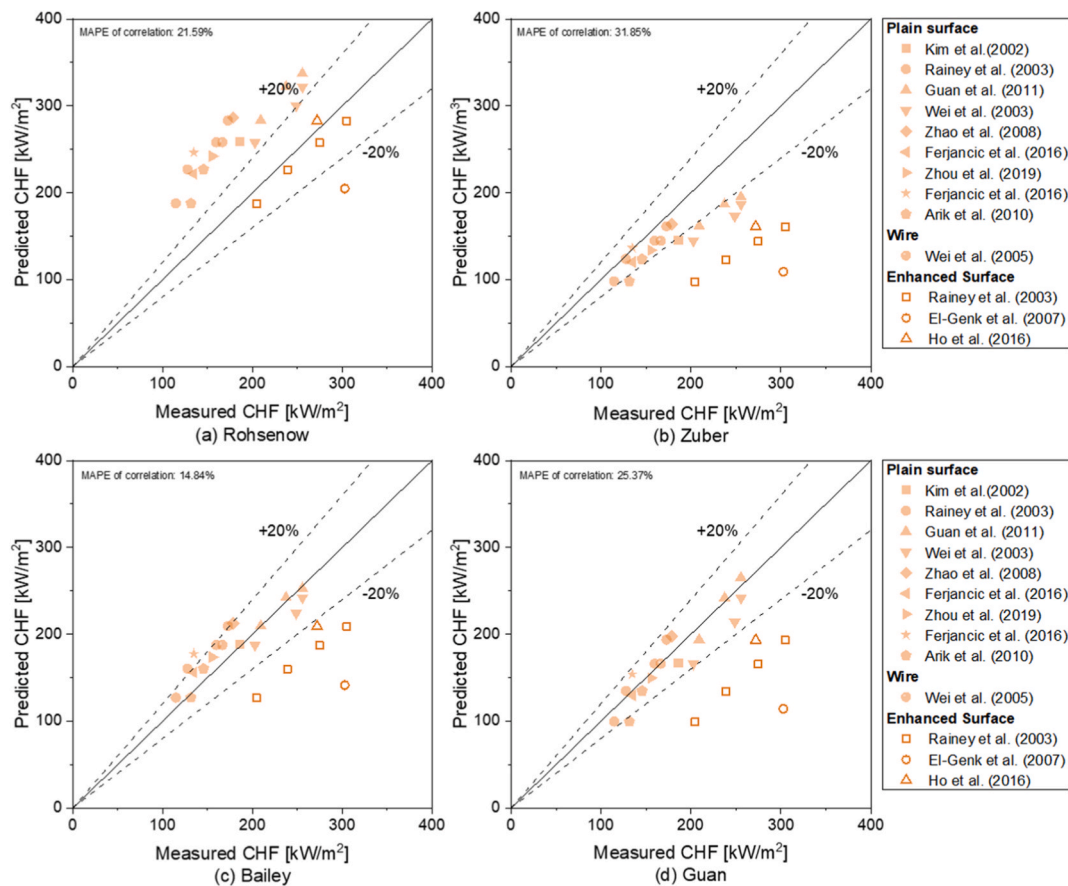


Fig. 12. Measured versus predicted CHF for FC-72 as a function of reduced pressure (Pred): compared across representative correlations.

been observed, the relative contribution of subcooling itself remains less clearly quantified compared to water.

Raj et al. [123] verified the applicability of a gravity-scaling parameter through variable-gravity and subcooled pool-boiling experiments with n-perfluorohexane/FC-72 mixtures, reporting that CHF increased with liquid subcooling. Kim and Benton [120] employed a microscale heater array to measure FC-72 boiling under low, normal, and high-gravity conditions, and found that CHF increased with both gravity and subcooling, with subcooling exerting a particularly strong effect. Rainey et al. [67] examined microporous finned surfaces over 30–150 kPa and 0–50 K subcooling, confirming a pronounced positive

dependence of CHF on subcooling. Wei and Honda [134] observed an almost linear increase in CHF with subcooling up to 45 K on micro-finned silicon chips under atmospheric pressure.

Further systematic studies have quantified subcooling coefficients across different surface types. Parker and El-Genk [128] demonstrated a linear rise of CHF with subcooling on porous graphite and plain copper, while El-Genk and Parker [130] extended this comparison to FC-72 and HFE-7100, confirming consistent linear behavior across inclination angles. O'Connor et al. [44] reported that CHF increased with “gassy subcooling” levels and that surface treatment enhanced subcooling sensitivity. More recently, Wang et al. [122] revisited Raj’s scaling

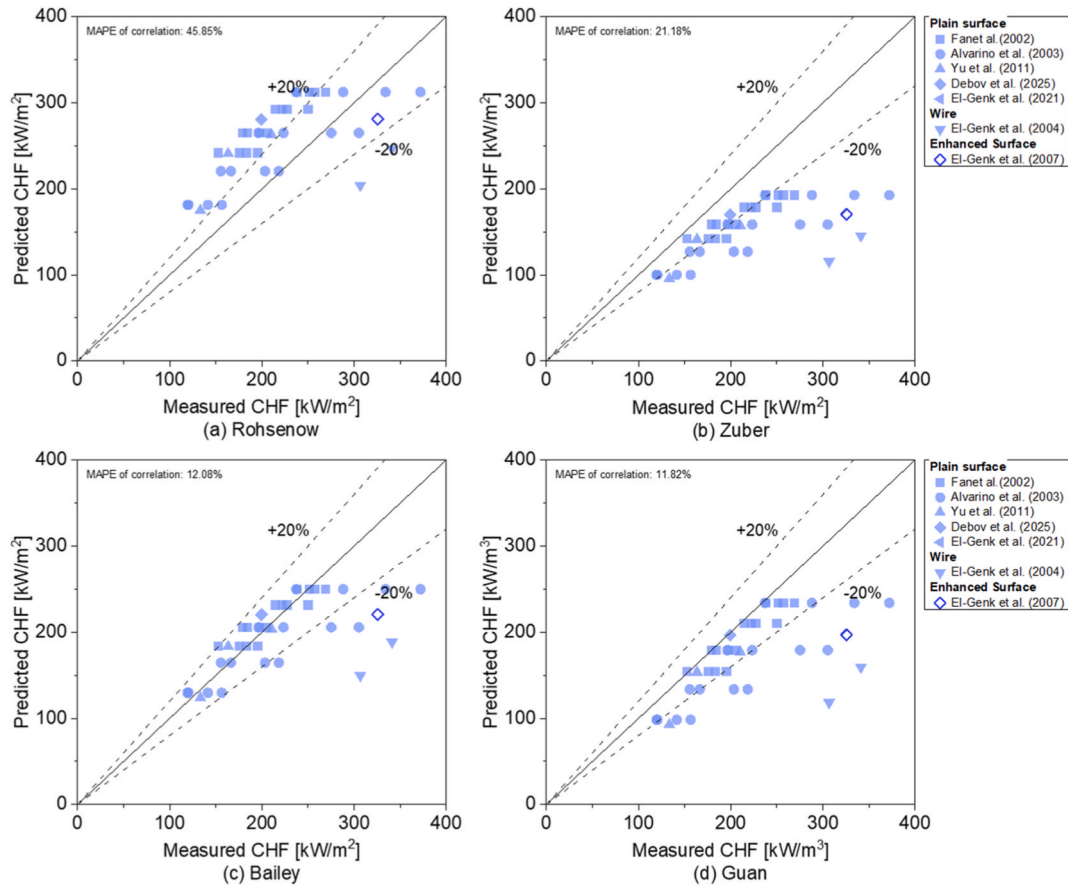


Fig. 13. Measured versus predicted CHF for HFE-7100 as a function of reduced pressure (Pred): compared across representative correlations.

Table 8

MAE and MAPE of representative CHF correlations for HFE-7100 on plate-type heaters.

	Rohsenow	Zuber	Bailey	Guan
MAE [kW/m ²]	0.0796	0.0454	0.0241	0.0254
MAPE [%]	45.8507	21.1758	12.0780	11.8193

Table 9

MAE and MAPE of representative CHF correlations for FC-72 on plate-type heaters.

	Rohsenow	Zuber	Bailey	Guan
MAE [kW/m ²]	0.0414	0.0756	0.0379	0.0584
MAPE [%]	21.5930	31.8451	14.8379	25.3650

approach in microgravity, reaffirming that CHF increases with subcooling and showing that the gravity-related power-law coefficient (mBDB) itself grows with the degree of subcooling.

A comparable analysis for FC-72 in Fig. 15 shows an opposite trend compared to the case of water: while correlations calibrated for water fail to predict FC-72 behavior, those specifically fitted to dielectric fluids such as Mudawar's model show improved consistency. This discrepancy can be rationalized by differences in thermophysical properties and bubble dynamics. FC-72 and HFE-7100, as discussed in Section 2, exhibit lower thermal diffusivity and latent heat, coupled with easier bubble detachment. While bubble departure time is quicker, the associated heat transport is smaller, and weaker thermal diffusion near the heated surface diminishes the sensitivity of CHF to subcooling.

6. Pressure and subcooling effects on HTC

6.1. Pressure dependence of HTC: water vs. dielectric fluids

In this section, the Rohsenow [144], Kutateladze [145], and Cooper [146] HTC correlations, which incorporate the influence of pressure, were compared with pressure dependent HTC data from the previous studies. The comparative results for different working fluids in terms of heat flux are presented in Figs. 16–18. Symbols represent experimental data reported in the literature, while solid lines denote the corresponding predictions obtained using the Rohsenow correlation with surface–fluid-dependent parameters adopted from previous studies. The results illustrate the pressure-dependent behavior of HTC and the varying predictive performance of the Rohsenow correlation across different working fluids.

For a more quantitative assessment of correlation performance, the predicted HTC values are plotted against the measured HTC values in Fig. 19, where all datasets are reorganized in a unified manner.

The difference in thermophysical properties between water and dielectric liquids, result in water having significantly larger heat flux and HTC values than dielectric liquids. These characteristics are reflected in the HTC correlations, through both empirical constants and fluid thermophysical values within the equations. However, as the methodology and magnitude to which these characteristics are incorporated differ, each correlation exhibits distinct results as well described in Fig. 19. Here, Symbols denote experimental data from the literature, while the solid diagonal line indicates perfect agreement. The dashed lines represent $\pm 20\%$ deviation bands. Therefore, there is a need to clarify which type of correlation is appropriate for a certain type of fluid. For detailed and quantitative comparison, error metrics including the mean absolute error (MAE) and mean absolute percentage error (MAPE)

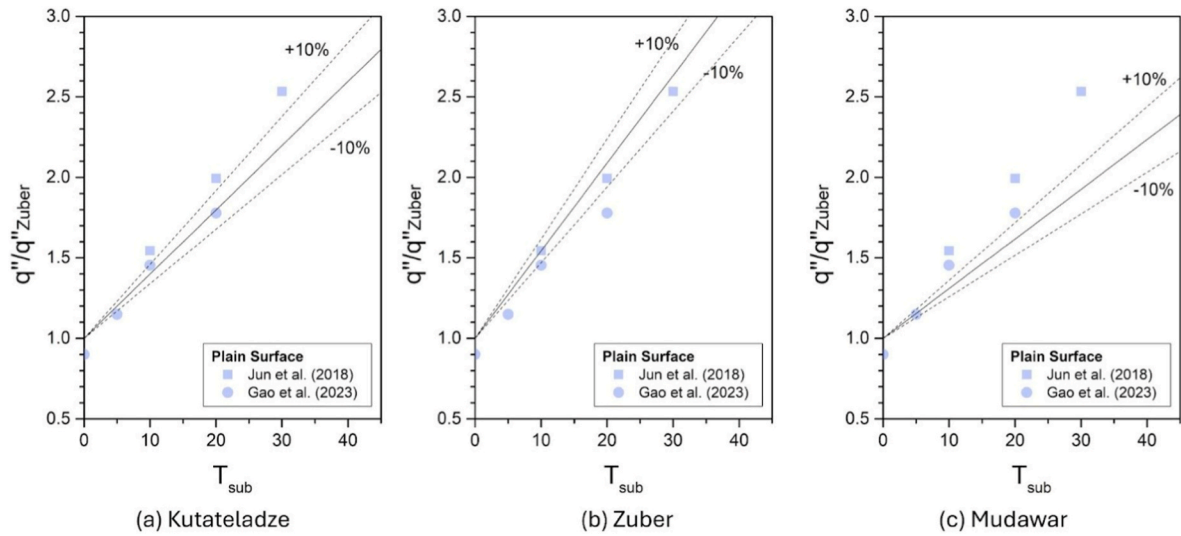


Fig. 14. Effect of subcooling on CHF for water: comparison of experimental data with representative correlations ((a) Kutateladze, (b) Zuber, (c) Mudawar).

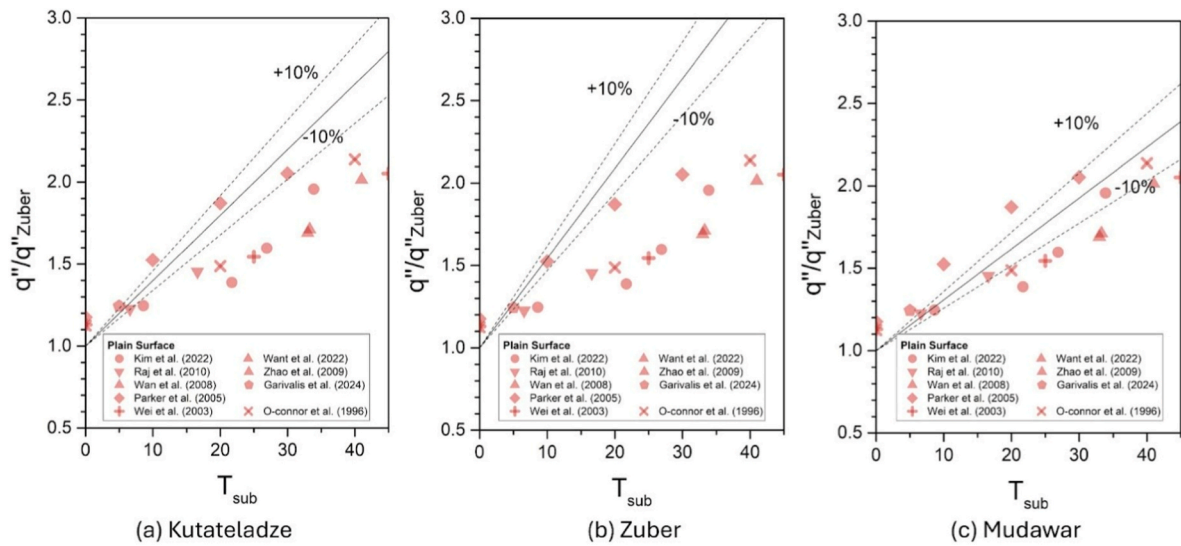


Fig. 15. Effect of subcooling on CHF for FC-72: comparison of experimental data with representative correlations ((a) Kutateladze, (b) Zuber, (c) Mudawar).

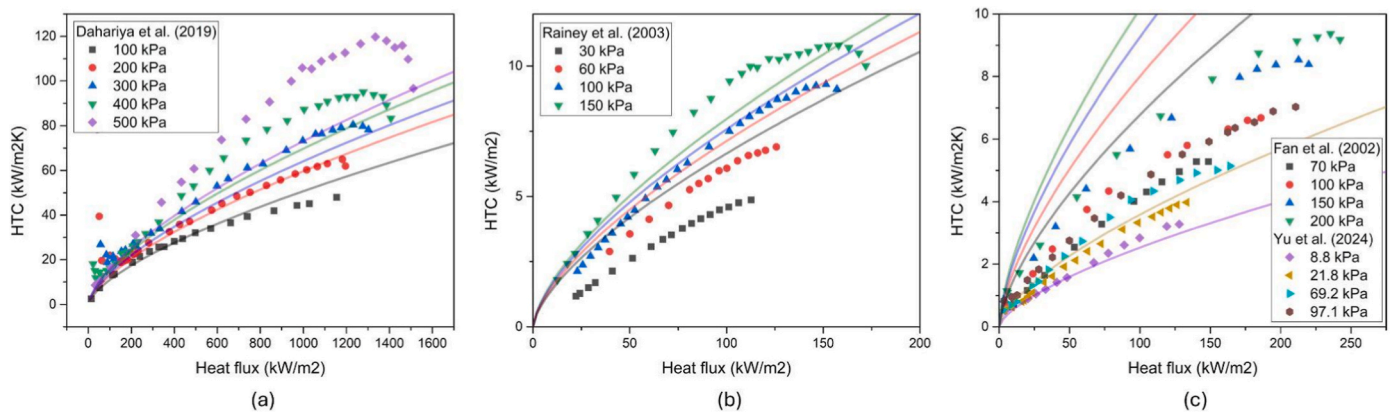


Fig. 16. Comparison of measured and predicted HTC using the Rohsenow correlation for (a) water, (b) HFE-7100, and (c) FC-72.

are summarized in Table 10 for each fluid with different pressure conditions.

The Rohsenow correlation is derived from vapor bubble and micro-

layer dynamics. It includes fluid properties evaluated at saturation conditions, such as viscosity, surface tension, fluid-gas density difference, latent heat of vaporization, and the Prandtl number. It also relies

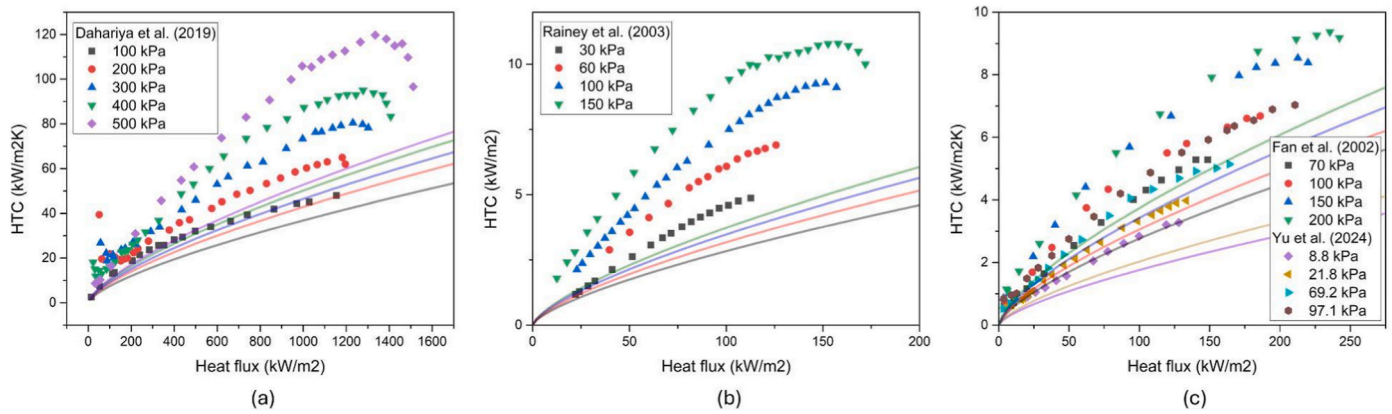


Fig. 17. Comparison of measured and predicted HTC using the Kutateladze correlation for (a) water, (b) HFE-7100, and (c) FC-72.

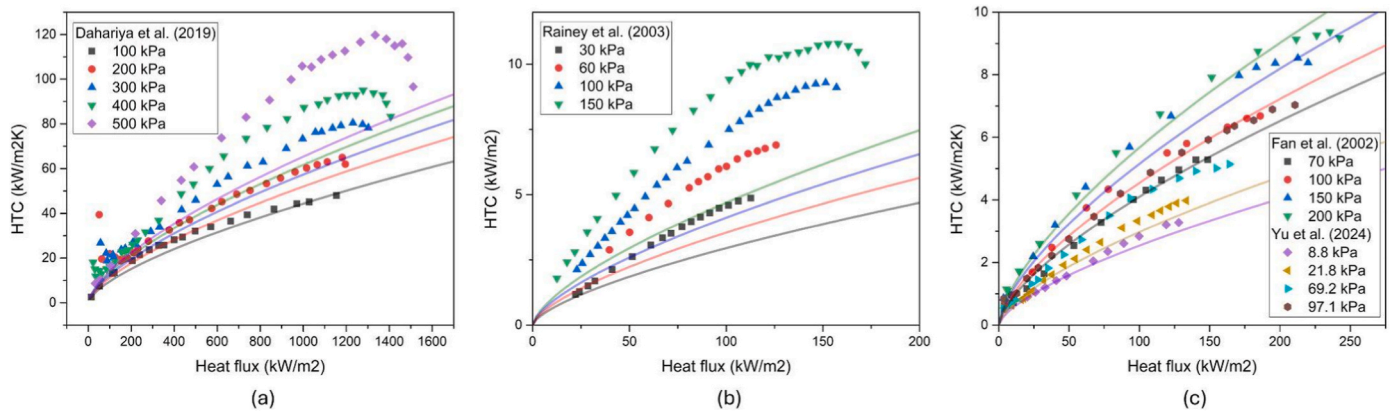


Fig. 18. Comparison of measured and predicted HTC using the Cooper correlation for (a) water, (b) HFE-7100, and (c) FC-72.

on an empirical constant C_{sf} to reflect the influence of surface–fluid combinations [147]. Therefore, the Rohsenow correlation can incorporate surface and fluid effects explicitly, and is relatively accurate when calibrated with experimental data. In this paper, the C_{sf} value was chosen based on surface–fluid combinations data from the previous studies, and predictions derived from the resulting correlation was the relatively most accurate for water and FC-72, among the three HTC correlations.

In contrast, the Kutateladze correlation is derived from dimensional analysis and similarity criteria of the heat transfer process [148]. It expresses boiling heat transfer through nondimensional groups, such as the Prandtl number, density ratio, and Laplace coefficient. The Kutateladze correlation is based on physical principles and does not use surface–fluid dependent empirical constants, making it advantageous for general boiling behavior analysis. However, as it does not reflect surface conditions such as roughness and wettability, its accuracy is often lower compared to correlations that are calibrated with experimental data. The predictions derived from the Kutateladze correlation in this paper, exhibited the worst accuracy among the three used correlations.

The Cooper correlation is an empirical correlation derived from a large set of experimental data covering different fluids, surfaces, and pressures. It is dependent on the molecular weight of the fluid, surface roughness and system pressure. It reflects the pressure effect through a reduced pressure term and logarithmic correction [149]. While the Cooper correlation does not reflect surface characteristics as explicitly as the Rohsenow correlation, it exhibited decent prediction performance across different fluid types, and especially HFE-7100, making it a practical choice when precise calibration conditions are not available. However, it is highly volatile depending on the surface roughness and may face restrictions in its predictions if the exact surface roughness

value is not available.

For water, the graphs indicate that the Rohsenow correlation provides the most accurate prediction of HTC. In contrast, for HFE-7100, significant deviations appear in the Rohsenow correlation at higher pressures, while the Cooper correlation aligns more closely with the experimental data. This difference may arise from how the two correlations incorporate fluid properties. The Rohsenow correlation explicitly uses fluid thermophysical properties in its equation, making its predictions sensitive to variations. By comparison, the Cooper correlation accounts for fluid properties through molecular weight, and also directly incorporates system pressure, which likely explains its reduced deviation with increasing pressure. Consequently, for pressure dependent HTC results of HFE-7100, the Cooper correlation offers greater accuracy than the Rohsenow equation, of which direct reliance on fluid thermo-physical properties may lead to larger errors under varying pressure conditions.

6.2. Subcooling dependence of HTC: water vs. dielectric fluids

Compared to CHF, systematic investigations of the effect of liquid subcooling on HTC are surprisingly limited, even for water. This reflects the historical focus on saturated boiling applications, where subcooling was regarded as a secondary effect. Across the limited available datasets, a general consensus emerges that while subcooling consistently increases CHF, its influence on HTC is much less pronounced, of fluid.

Several representative studies illustrate this point. Jun et al. [75], Watwe et al. [76], and Rainey et al. [77], concluded that in the fully developed nucleate boiling regime, liquid subcooling has little to no measurable impact on HTC. This insensitivity arises because counter-acting mechanisms—such as reduced bubble departure diameter and

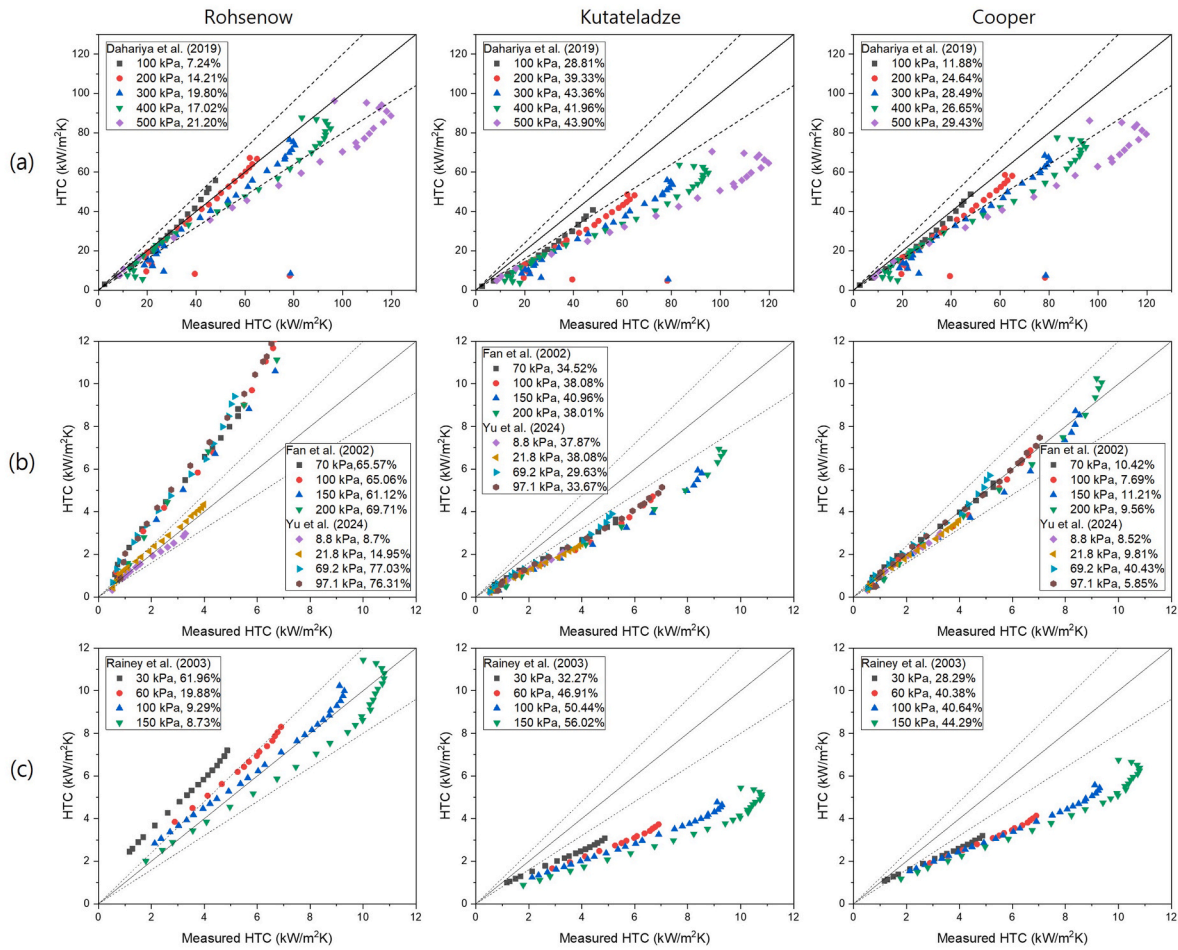


Fig. 19. Predicted versus measured heat transfer coefficients (HTC) for (a) water, (b) HFE-7100, and (c) FC-72 using the Rohsenow (left column), Kutateladze (middle column), and Cooper (right column) correlations under various pressure conditions, and their respective MAPE.

Table 10

Summary of MAE [$\text{kW}/\text{m}^2\text{K}$] and MAPE [%] for HTC predictions of the evaluated correlations for each working fluid under various pressure conditions.

	Rohsenow [144]		Kutateladze [145]		Cooper [146]	
Water	MAE	MAPE	MAE	MAPE	MAE	MAPE
100 kPa [Dahariya et al. (2019)]	2.0817	7.2401	7.3278	28.8148	2.7899	11.8762
200 kPa [Dahariya et al. (2019)]	5.8516	14.2059	15.6796	39.3314	9.8830	24.6440
300 kPa [Dahariya et al. (2019)]	9.0769	19.7987	20.9023	43.3585	13.5307	28.4923
400 kPa [Dahariya et al. (2019)]	8.2067	17.0243	21.1568	41.9619	13.3264	26.6496
500 kPa [Dahariya et al. (2019)]	19.5013	21.1988	37.2076	43.9031	26.2441	29.4319
HFE-7100	MAE	MAPE	MAE	MAPE	MAE	MAPE
70 kPa [Fan et al. (2002)]	1.8632	65.5675	0.9398	34.5232	0.1361	10.4219
100 kPa [Fan et al. (2002)]	2.8060	65.0645	1.4293	38.0817	0.2133	7.6856
150 kPa [Fan et al. (2002)]	3.7002	61.1248	2.0372	40.9608	0.4204	11.2065
200 kPa [Fan et al. (2002)]	4.6043	69.7066	1.9769	38.0136	0.4132	9.5595
8.8 kPa [Yu et al. (2024)]	0.1574	8.7006	0.7062	37.8657	0.1519	8.5217
21.8 kPa [Yu et al. (2024)]	0.2446	14.9513	0.8528	38.0829	0.2161	9.8111
69.2 kPa [Yu et al. (2024)]	2.1586	77.0250	0.8390	29.6330	0.2031	10.4260
97.1 kPa [Yu et al. (2024)]	3.1440	76.3100	1.2641	33.6684	0.1303	5.8453
FC-72	MAE	MAPE	MAE	MAPE	MAE	MAPE
30 kPa [Rainey et al. (2003)]	1.7658	61.9603	1.1354	32.2672	1.0262	28.2862
60 kPa [Rainey et al. (2003)]	1.0470	19.8842	2.5944	46.9140	2.2443	40.3838
100 kPa [Rainey et al. (2003)]	0.4067	9.2901	3.1997	50.4393	2.6257	40.6445
150 kPa [Rainey et al. (2003)]	0.7143	8.7338	4.5411	56.0206	3.6246	44.2909

frequency versus thinner superheated liquid layers—tend to balance one another, leading to collapse of boiling curves onto a single trend irrespective of subcooling level. By contrast, El-Genk [130] reported that

HTC decreases with increasing subcooling in FC-72 and HFE-7100 on both porous graphite and plain copper surfaces, attributing this trend to higher wall superheat which reduces the ratio of heat flux to

temperature difference ($h = q/\Delta T$). Gao et al. [137] presented a more nuanced picture: on micro-pillar surfaces, subcooling enhanced nucleate boiling at higher heat fluxes by activating additional nucleation sites and suppressing bubble coalescence, although at low heat fluxes the effect was mainly to extend the natural convection regime. Sakurai and Fukuda [78], while not focused directly on HTC, also demonstrated that CHF mechanisms under subcooling (hydrodynamic instability vs. heterogeneous spontaneous nucleation) evolve nonlinearly with pressure, indirectly suggesting that HTC behavior may similarly depend on regime transitions.

Taken together, existing studies indicate that the influence of liquid subcooling on HTC in pool boiling is generally modest and highly context dependent. For water, subcooling-induced HTC enhancement is primarily observed at low heat fluxes, where an increased temperature difference between the heated surface and the bulk liquid strengthens natural convection and Marangoni-driven flow [67,76]. These effects, together with stabilized bubble nucleation and delayed vapor blanket formation, promote more efficient liquid replenishment near the heated surface and thereby enhance heat transfer, as also discussed in the flow boiling literature [150,151]. However, once nucleate boiling becomes fully developed, counteracting mechanisms largely offset one another and the sensitivity of HTC to subcooling becomes weak or negligible. These regime-dependent trends are summarized schematically in Fig. 20, which highlights the transition from convection-assisted heat transfer at low heat fluxes to a fully developed nucleate boiling regime where competing mechanisms diminish the net influence of subcooling on HTC.

For dielectric fluids such as FC-72 and HFE-7100, several studies suggest a potentially stronger dependence on subcooling, attributed to suppressed early bubble coalescence and enhanced liquid replenishment near the heated surface. Nevertheless, available datasets remain sparse and are frequently confounded by concurrent variations in pressure, heater geometry, surface structure, or gravity level, making it difficult to isolate the independent contribution of subcooling to HTC behavior.

As a result, HTC correlations that explicitly incorporate subcooling effects are rare. Most existing formulations account for subcooling only implicitly, through empirical constants or secondary correction terms within broader boiling heat transfer models. Consequently, no clear consensus has yet emerged regarding the predictive accuracy of different HTC correlations under subcooled conditions. Expanding high-quality, systematically controlled experimental datasets for both water and dielectric fluids is therefore essential for improving HTC model validation and refinement.

6.3. Summary of recommended correlations

Based on the critical assessment of experimental data and predictive models presented in Sections 5 and 6, this study identifies the most suitable correlations for predicting CHF and HTC under various operating conditions. Table 11 summarizes these recommendations, categorized by fluid type, pressure range, and subcooling conditions. The

table also highlights key remarks regarding empirical constants and application parameters essential for practical implementation.

As summarized in Table 11, the prediction accuracy depends critically on how pressure, wettability, and fluid properties are integrated. For CHF, classical hydrodynamic instability models like Zuber's serve as a valid baseline for water at low pressures but significantly underpredict performance at elevated pressures due to wettability variations. In such high-pressure regimes, surface-sensitive models like Rajvanshi or Kumada are preferred. Conversely, for dielectric fluids, the Bailey correlation provides the most robust CHF predictions across varying pressures by effectively addressing the low surface tension characteristics. Under subcooled conditions for dielectric fluids, the Mudawar correlation is recommended as it explicitly accounts for the sensible heat contribution via the Jakob number.

Regarding HTC, the choice involves a trade-off between calibration requirements and generality; the Rohsenow correlation offers high accuracy for water and FC-72 but requires surface-specific empirical constants C_{sf} , whereas the Cooper correlation provides a robust alternative for Novec-7100 by leveraging reduced pressure and molecular weight. Finally, while subcooling significantly enhances CHF, its effect on HTC is minimal in the fully developed nucleate boiling regime, suggesting that saturated HTC correlations can be applied with reasonable accuracy, even under subcooled conditions.

7. Techno-economic implications of pressure and subcooling effects

7.1. Framework for techno-economic and energy-based cost assessment

A comprehensive techno-economic analysis (TEA) or life-cycle assessment (LCA) of pressurized and subcooled immersion cooling systems is essential for system design and optimization, as it requires detailed system layouts, component specifications, and vendor-dependent pricing data. Such information is inherently application-specific and, moreover, detailed cost quotations are not extensively reported in the open literature. Nevertheless, the economic and sustainability implications of enhanced boiling performance can be meaningfully assessed using a structured, system-level cost framework and representative cost parameters, following the methodology adopted in recent economic analyses of immersion-cooled data centers [97,152].

Kanbur et al. [97] conducted long-term cost assessments based on experimental data and, in addition, reported thermoeconomic indicators (including exergy-based assessments) to quantify system-level economic losses. It should be noted, however, that their experiments were conducted at laboratory scale, and practical design constraints limited the achievable heat dissipation capacity and system power level. Despite these limitations, their study remains one of the few publicly available and valuable references that provides a transparent and experimentally grounded cost-accounting framework for immersion cooling systems.

In the present study, their cost-accounting methodology and key input parameters are adopted, and monetary values are converted to

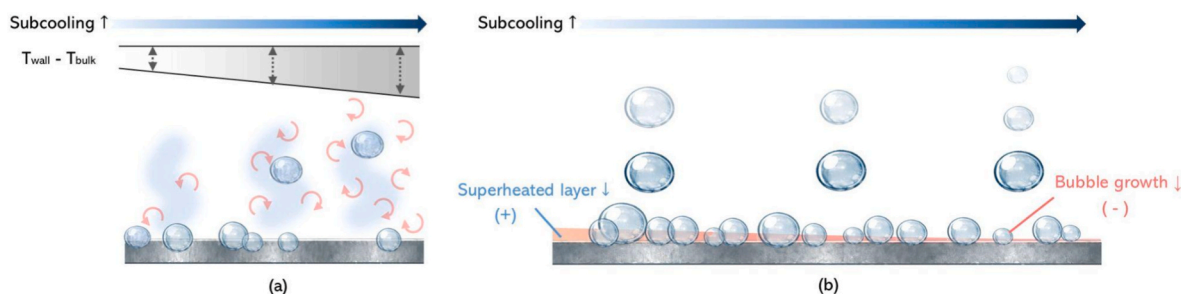


Fig. 20. Conceptual schematic illustrating the regime-dependent influence of liquid subcooling on pool boiling heat transfer at (a) low heat flux and (b) fully developed nucleate boiling.

Table 11
Summary of the correlations under various conditions.

Parameter	Fluid Type	Operating Condition	Recommended Correlation	Comments /Remarks
CHF	Water	Low Pressure	Zuber	Serves as a valid baseline for standard conditions (typically using coefficient $K \sim 0.131$), though it underpredicts at elevated pressures.
		High Pressure	Kumada	Recommended for high-pressure regimes as they account for pressure-dependent wettability enhancement, which classical models fail to capture.
		Varying Pressure	Bailey	Exhibits the closest overall agreement across the reduced-pressure range by correcting Zuber-type scaling for low surface tension fluids.
Dielectric Fluids	Dielectric Fluids	Subcooled Conditions	Mudawar	Specifically fitted for dielectric fluids using the Jakob number; demonstrates better consistency than correlations calibrated for water.
		Varying Pressure	Rohsenow	Provided the most accurate predictions among tested models, provided that the empirical constant (C_{sf}) is calibrated for the specific surface-fluid combination.
		Varying Pressure	Cooper	Aligns closer with experimental data than Rohsenow for HFE-7100 by explicitly incorporating reduced pressure and molecular weight.
HTC	Water & FC-72	Varying Pressure	Rohsenow	Provided the most accurate predictions among tested models, provided that the empirical constant (C_{sf}) is calibrated for the specific surface-fluid combination.
	HFE-7100	Varying Pressure	Cooper	Aligns closer with experimental data than Rohsenow for HFE-7100 by explicitly incorporating reduced pressure and molecular weight.

Korean Won (KRW) using the current exchange rate [153]. Consistent with the economic analysis based on the present worth (PW) method presented in Ref. [152], the total cost is expressed in terms of cumulative present worth (CPW) and decomposed into fixed capital investment (FCI) and present worth of cumulative operational expenditures (PW_{OPEX}):

$$CPW = FCI + PW_{OPEX} \quad (13)$$

Capital costs are associated with the procurement and installation of major equipment, whereas operational costs are dominated by

electricity consumption, carbon-related charges, operation and maintenance, and replacement of consumables.

Following conventional practice, the FCI consists of numerous cost components that are situation-dependent and difficult to track individually. Accordingly, it is commonly expressed as a multiple of the purchased equipment cost (PEC):

$$FCI = 4.30 \times PEC \quad (14)$$

Estimation factors have also been reported in the literature, with values spanning a broad range, typically between 3 and 10 [154–156]. In practice, however, commonly adopted factors tend to lie in the range of 4–5 [157]. Accordingly, an intermediate value of 4.5 is adopted in the present study. The PEC is taken directly from the reference study [97] and converted to KRW using the exchange rate in Ref. [153], yielding a representative value of approximately 452,697,026 KRW.

Operational expenditures are primarily driven by annual energy consumption and can be written as

$$C_{OPEX} = C_{energy} + C_{carbon} + C_{OM} + C_R \quad (15)$$

where C_{OPEX} denotes the annual operating cost, C_{energy} denotes electricity cost associated with auxiliary components (pumps, fans, chillers), C_{carbon} represents carbon emission costs linked directly to electricity consumption, C_{OM} accounts for operation and maintenance, and C_R represents replacement costs related to working fluid loss or component degradation. Importantly, the first two terms are directly proportional to cooling-system power consumption and therefore provide a direct quantitative bridge between boiling heat transfer performance and sustainability metrics.

To enable a consistent comparison on an annual basis, the cumulative operational expenditure is converted into an equivalent C_{OPEX} using the capital recovery factor (CRF):

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (16)$$

$$PW_{OPEX} = C_{OPEX} / CRF \quad (17)$$

where i is the discount rate (set as 2.0% in this study) and n is the system lifetime. In line with previous studies [97,152], the individual operational cost components are evaluated as follows:

- (i) **Energy cost** (C_{energy}), calculated from the total electricity consumption of the cooling system and expressed as

$$C_{energy} = \dot{Q}_S c_{elec} \tau, PW_{energy} = \frac{C_{energy}}{CRF} \quad (18)$$

where $\dot{Q}_S$ is the total electrical power consumption of the cooling system, including circulation pumps and dry-cooler fans (kW), $c_{elec} = 154.848$ (KRW/kWh) is the unit electricity price under Korean conditions [158], and $\tau = 8760h$ is the annual operating time.

- (ii) **Carbon emission cost** (C_{carbon}), estimated from electricity consumption combined with the grid-specific emission factor and applicable carbon pricing, given by

$$C_{carbon} = \dot{Q}_S \tau CDE \frac{c_{carbon}}{1000}, PW_{carbon} = \frac{C_{carbon}}{CRF} \quad (19)$$

where $CDE = 0.4113 \text{ kg CO}_2/\text{kWh}$ is the Korean grid CO_2 emission factor [159], $c_{carbon} = 9000$ KRW/ tCO_2 is the representative carbon price [160], and the factor $1/1000$ converts kilograms of CO_2 to tonnes [159].

- (iii) **Operation and maintenance cost** (C_{OM}), assumed as a fixed fraction of the purchased equipment cost (PEC) following previous studies [97,152,154], and calculated as

$$C_{OM} = PEC \cdot r_{OM}, PW_{OM} = \frac{C_{OM}}{CRF} \quad (20)$$

where r_{OM} is the operation and maintenance cost ratio, taken as 1.092%.

- (iv) **Replacement cost (C_R)**, associated with dielectric liquid loss due to evaporation or degradation and the replacement of related auxiliary components. While this term is system- and operation-dependent, it is adopted from the baseline system reported in Ref. [97] for consistency. In Ref. [152], the annual replacement cost was reported as $C_R = 5000$ SGD, corresponding to 5,604,420 KRW when converted using a representative exchange rate of 1 SGD = 1120 KRW [153].

$$PW_R = \frac{C_R}{CRF} \quad (21)$$

Fig. 21 presents a representative decomposition of the cumulative present worth over system lifetime reconstructed based on the cost-accounting framework above [97]. As shown, operational expenditures accumulate over time and constitute an increasingly important fraction of the total cost. Among the operational cost components, the present worth of replacement cost (PW_R) and energy cost (PW_{energy}) accounts for a noticeable share of the total operating expenditure. The contribution of carbon emission cost (PW_{carbon}) is found to be comparatively small compared to other costs.

It should be emphasized, however, that these results are based on laboratory-scale experimental conditions, under which the replacement cost may be overestimated due to non-optimized operating procedures and scale-related inefficiencies. Moreover, the relative magnitude of individual cost components can vary significantly depending on system scale, design choices, and operating conditions. Accordingly, the cost breakdown shown in Fig. 21 should be interpreted as a reference illustration rather than a definitive economic assessment.

For reference, Table 12 summarizes the corresponding numerical values of the fixed capital investment and operational cost components

Table 12

Cost-over-time summary for a 20-year system lifetime under $\dot{Q}_S = 0.8$ kW.

Unit: 10^8 [KRW]			
FCI		PW_{OPEX}	
Item	Cost [10^8 KRW]	Item	Cost [10^8 KRW]
Server Tank	0.314	C_{energy}	0.011
Pump	0.011	C_{carbon}	0.00026
Dry Tower	0.359	C_{OM}	0.011
Condenser	0.269	C_R	0.056
Dielectric Liquid	0.054		
PEC	1.006	C_{OPEX}	0.078
FCI	4.527	PW_{OPEX}	1.278

for the $\dot{Q}_S = 0.8$ kW over a 20-year system lifetime, reconstructed from the cost-accounting framework of [97].

7.2. Quantitative impact of HTC and CHF enhancement on power consumption and operating cost

Based on the cost-accounting framework introduced in Section 7.1, this section examines how pressure- and subcooling-induced enhancements in boiling heat transfer performance influence auxiliary power consumption and operating costs. A fully quantitative techno-economic optimization is not attempted, as immersion cooling systems exhibit wide variability in design configurations, component selection, and cost structures. Instead, a qualitative system-level analysis is combined with simplified, order-of-magnitude estimates to elucidate the dominant cost-driving mechanisms.

Fig. 22 provides a conceptual overview of the competing effects associated with pressure and subcooling enhancement, highlighting the trade-off between increased capital investment and reduced operating expenditure.

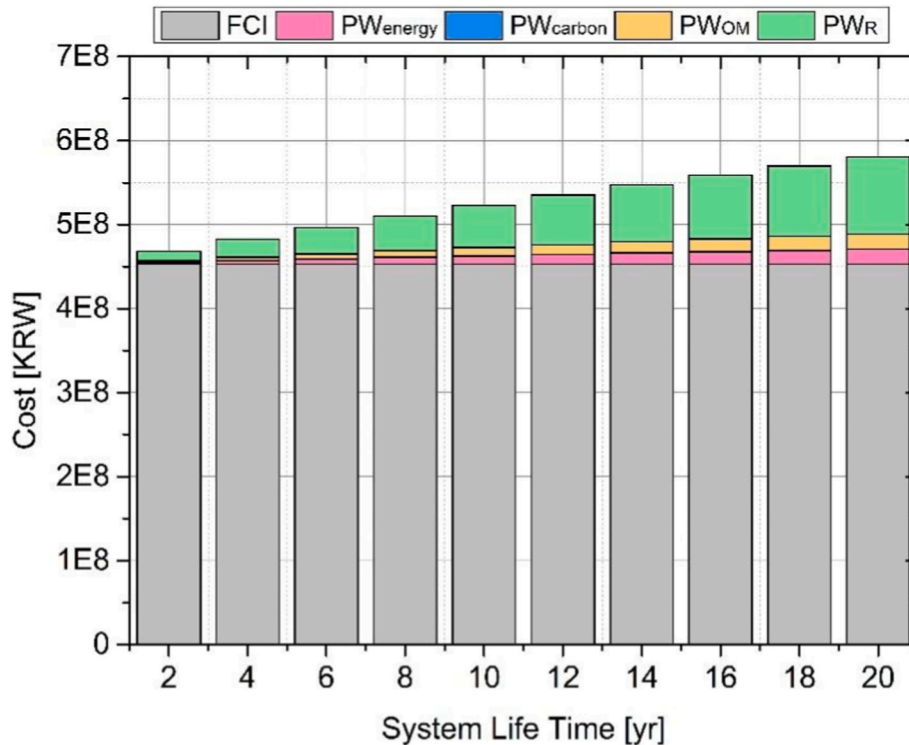


Fig. 21. Breakdown of cumulative present worth (CPW) cost components over system lifetime for a representative immersion cooling system (based on previous lab-scale experiment [97]).

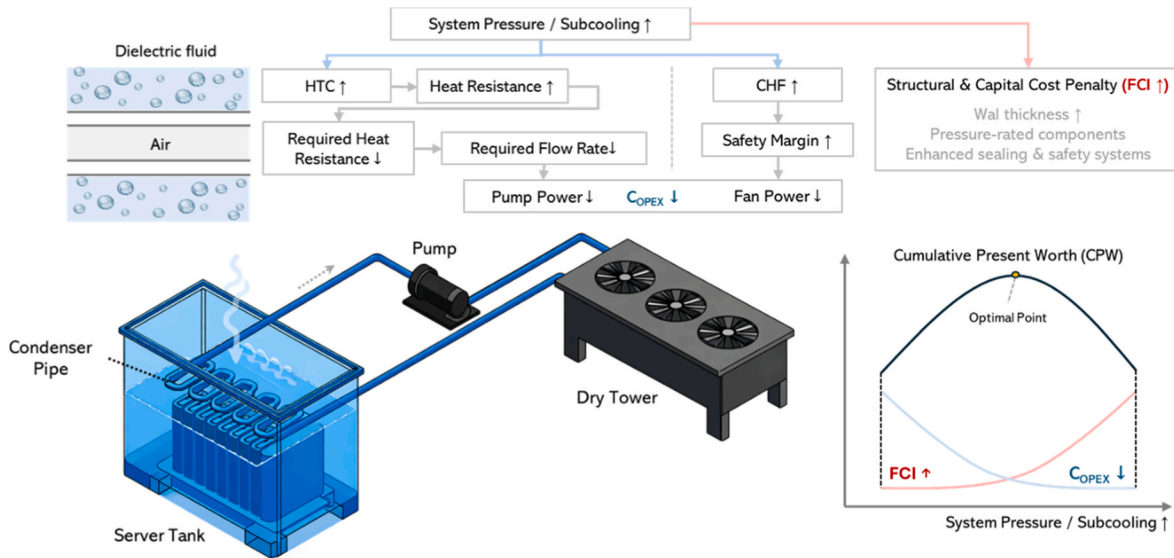


Fig. 22. Conceptual illustration of system-level cost trade-offs associated with pressure and subcooling enhancement in immersion cooling systems. (The schematic of two-phase immersion cooling system was reproduced from previous literature [97].)

7.2.1. HTC enhancement and pumping power reduction

For a given thermal load, the required heat removal is fixed, while the effective heat transfer area and allowable temperature difference are constrained by packaging density and device reliability. Under these constraints, enhancement of HTC improves local heat removal efficiency at the device surface.

Although boiling occurs under pool-boiling conditions, heat transport at the system level relies on forced circulation of the coolant through the bulk liquid and external heat rejection loop. Consequently, auxiliary power consumption is dominated by pumps, piping, and heat exchangers rather than by the boiling process itself. From a thermal-resistance perspective, increased HTC reduces the interfacial heat transfer resistance, enabling the same heat load to be removed at a lower coolant flow rate. Reduced flow rates lead to lower pressure losses and, therefore, significantly reduced pumping power, owing to the nonlinear dependence of pumping power on flow rate under turbulent conditions.

As a result, even moderate HTC enhancement—achieved through elevated pressure or increased liquid subcooling—can yield disproportionately large reductions in pumping power, directly lowering operating cost and associated carbon emissions.

7.2.2. CHF enhancement and fan power reduction

While HTC enhancement primarily affects pumping power, improvements in CHF influence energy consumption through design constraints. CHF sets the maximum allowable local heat flux and therefore limits the minimum heat transfer area required for safe operation.

An increase in CHF relaxes this constraint and provides additional design flexibility. Rather than reducing the physical size of heat rejection components, this margin can be exploited to lower airflow rates in air-cooled condensers or dry coolers, where thermal resistance is often dominated by the air side. Because fan power scales strongly with airflow rate, even modest airflow reductions can result in substantial fan power savings. In this way, CHF enhancement indirectly contributes to reduced auxiliary power consumption and long-term operating cost.

7.2.3. Cost implications of pressure and subcooling enhancement

Enhancing HTC and CHF through increased pressure or subcooling inevitably introduces design trade-offs. Higher operating pressure requires pressure-rated components, including reinforced piping, valves, structural elements, and safety systems. These requirements primarily affect initial capital cost. In dielectric immersion cooling systems, however, feasible pressure increases are inherently limited by fluid

properties and safety constraints. As a result, only moderate pressure increments are considered realistic in this study, and the associated increase in initial capital cost is assumed to remain relatively modest, on the order of approximately 10%.

In contrast, operating expenditures (particularly those associated with auxiliary electrical consumption) exhibit a fundamentally different scaling behavior. Pumping and fan power are strongly dependent on flow rate, typically following nonlinear hydraulic relationships. Consequently, modest pressure-induced enhancements in HTC and CHF can relax required circulation margins and enable reductions in coolant or airflow rates, leading to disproportionately larger reductions in auxiliary power consumption. While capital cost responds approximately linearly to moderate structural modifications, operating cost may decrease nonlinearly due to cubic flow–power scaling.

Improved thermal performance may also reduce maintenance and replacement costs by lowering mechanical stress and operational intensity. However, rigorous quantification of reductions in C_{OM} and C_R would require detailed reliability modeling and system-specific validation. To avoid speculative assumptions, the present exploratory assessment conservatively considers reductions only in electricity-related cost components (C_{energy} and C_{carbon}).

Accordingly, the modified cumulative present worth is evaluated as:

$$CPW_{expected} = FCI(1 + \Delta) + \frac{[(C_{energy} + C_{carbon})(1 - \alpha) + C_{OM} + C_R]}{CRF} \quad (22)$$

where Δ denotes the assumed fractional increase in capital cost (up to 10%), and α represents the assumed fractional reduction in electricity-related operating costs.

Based on these considerations, an exploratory sensitivity analysis was conducted by varying auxiliary power reduction between 0 and 50% and capital cost increase between 0 and 10%. These ranges do not represent design predictions but instead reflecting the different sensitivity of structural cost to pressure (approximately proportional under moderate design modifications) and of auxiliary power consumption to flow rate, which follows established nonlinear hydraulic and aerodynamic scaling relationships. The resulting contour map in Fig. 23 illustrates that, beyond a threshold level of auxiliary power reduction, operational savings can offset moderate capital penalties, indicating the existence of economically favorable operating regions.

It is important to emphasize that this formulation does not arise from a fully coupled thermo-economic optimization or an experimentally

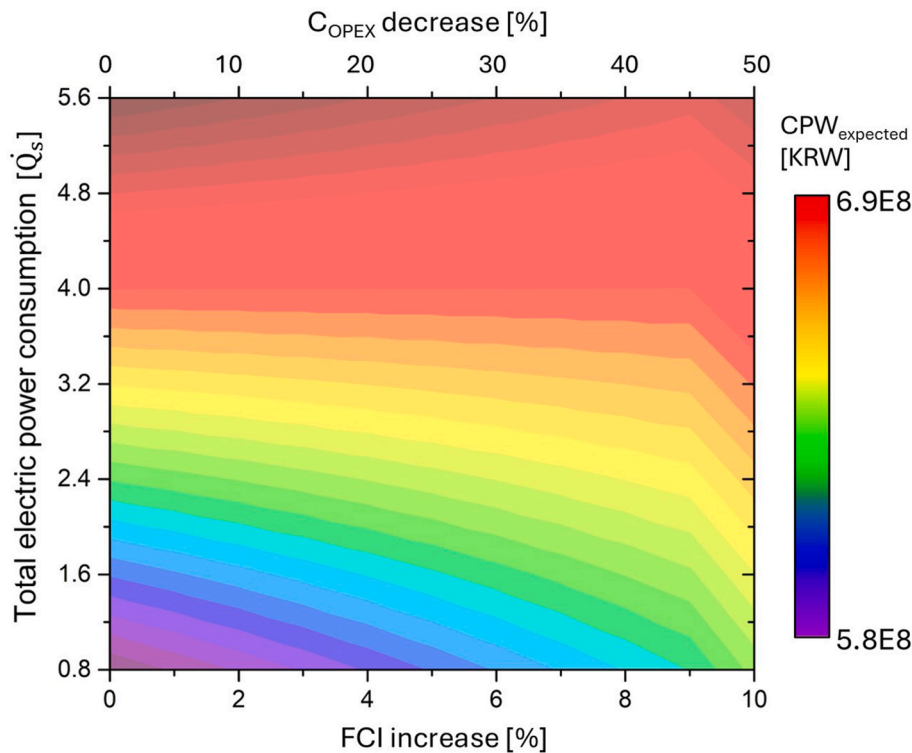


Fig. 23. Contour map showing the qualitative sensitivity of cumulative present worth to capital cost increase and auxiliary power reduction under exploratory assumptions.

validated pressure–cost correlation. The assumed parameter ranges were introduced solely to identify potential break-even regions through a simplified sensitivity framework. A rigorous quantitative mapping between pressure or subcooling enhancement and specific capital and operating cost variations would require experimentally validated heat-transfer correlations, detailed component-level sizing, and integrated system-level cost modeling, which are beyond the scope of the present study and remain subjects for future research.

8. Conclusion

This review has critically examined the influence of system pressure and liquid subcooling on the CHF and HTC in pool boiling, focusing on water and representative dielectric fluids such as FC-72 and HFE-7100. By restricting attention to plain surfaces and carefully controlled datasets, the analysis provides a consistent basis for comparing experiments with widely used predictive models and for identifying where current correlations remain reliable or systematically breakdown.

For water, CHF has been investigated extensively over a broad pressure range. Classical hydrodynamic-instability models, including Zuber- and Kutateladze-type formulations, provide valuable reference scaling, yet they tend to underpredict CHF at elevated pressures because they do not capture pressure-dependent changes in wettability. Mechanistic extensions incorporating surface-related effects, such as macro- or microlayer concepts, can improve agreement, but their dependence on specific surface conditions limits generality. For dielectric fluids, the available CHF database remains comparatively sparse, particularly at high pressures. Nonetheless, existing results consistently indicate CHF enhancement with increasing pressure, and Bailey's correlation offers the most robust agreement among representative models. This supports the view that Zuber-type scaling can serve as a broadly applicable framework, provided that appropriate corrections are introduced for low-surface-tension fluids.

Regarding subcooling, both water and dielectric fluids exhibit a clear increase in CHF as the bulk liquid temperature decreases below

saturation. The scarcity of systematic subcooling datasets for water likely reflects the historical emphasis on saturated boiling in many water-based applications. In contrast, dielectric-fluid immersion cooling commonly operates under undersaturated conditions, where external condensation and recovery loops naturally deliver subcooled liquid to the heated surface. This practical distinction helps explain why subcooling has been less emphasized in water studies but is central to dielectric-fluid applications. Under partial undersaturation, however, simultaneous nucleation and condensation complicate both experimentation and modeling, and the number of interacting variables increases substantially. This highlights that efforts to rigorously quantify these combined effects within unified correlations remain limited.

Across representative HTC correlations, Rohsenow's model yielded the best predictions for water and FC-72 when properly calibrated, Cooper's correlation was comparatively versatile for dielectric fluids owing to its reduced-pressure formulation, and Kutateladze's similarity-based approach was least accurate across the surveyed cases. For HTC specifically, subcooling plays a markedly smaller role than for CHF. Once nucleate boiling is established, HTC is governed primarily by nucleation site density and bubble dynamics, with only weak sensitivity to bulk subcooling.

Building on these findings, this review organizes and recommends classical CHF and HTC correlations for plain surfaces by fluid class, pressure range, and subcooling condition, providing a practical best-fit map for model selection. Although isolating individual factors is inherently difficult in boiling experiments, where multiple coupled variables simultaneously influence outcomes, this work adopts a dimensionless-group-based perspective to systematically organize the available evidence. Through this framework, the origins of divergence among existing correlations are clarified, the necessity of condition-dependent correlation selection is rationalized, and physically grounded insights are extracted regarding the respective roles of fluid properties, pressure, and subcooling in governing CHF and HTC behavior.

Finally, the review discusses the potential techno-economic implications of enhancing CHF and HTC at the system level. Using

experimental, system-based life-cycle assessment and techno-economic studies of two-phase immersion cooling as a reference point, we qualitatively examine where pressure- or subcooling-driven performance gains could increase or decrease cost, including auxiliary power, component sizing margins, operating constraints, and balance-of-plant complexity. We also outline the plausibility of an optimum operating sweet spot or a net cost-reduction pathway. Importantly, these conclusions are intentionally framed as qualitative. The present discussion does not constitute a full techno-economic analysis or life-cycle assessment, and quantitatively defensible optimization will require broader and more systematic high-quality datasets, especially for dielectric fluids at elevated pressures and for partially undersaturated conditions, together with targeted experiments and validated system-level models.

From a broader sustainability perspective, system pressure and liquid subcooling, considered individually or in combination with other operating parameters like surface modification, can be treated as effective design levers for improving thermal efficiency, reducing auxiliary energy consumption, and ultimately enhancing the sustainability of energy and cooling systems relevant to renewable and low-carbon technologies. However, fully realizing this potential requires overcoming several fundamental limitations in the current experimental database and modeling framework.

Accordingly, future research should prioritize the following directions:

- Expanding high quality CHF and HTC databases for dielectric fluids across a wide range of pressures and subcooling levels.
- Designing experiments that isolate subcooling effects while minimizing confounding variables.
- Developing unified models that integrate wettability, thermophysical property variations, and surface effects.

Addressing these priorities will not only improve the predictive reliability of pool boiling CHF and HTC correlations but also support the development of sustainable, high efficiency thermal management strategies in renewable energy systems, battery ESS, data centers, aerospace platforms, and other high power density applications.

Declaration of generative AI use

The authors acknowledge the limited use of ChatGPT for post-processing and visual enhancement of selected figures (Figs. 6 and 22). This tool was not used for data generation, analysis, or interpretation. All scientific content and conclusions were produced by the authors. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] Kim SJ, Choi YS, Jo YB, Park IW, Park H. A review of metal foam-enhanced pool boiling. *Renew Sustain Energy Rev* 2025 Mar;210:115176. <https://doi.org/10.1016/j.rser.2024.115176>.
- [2] Rashidi S, Hormozi F, Sarafraz MM. Potentials of boiling heat transfer in advanced thermal energy systems. *J Therm Anal Calorim* 2021 Feb 10;143(3):1833–54. <https://doi.org/10.1007/s10973-020-09511-3>.
- [3] Ni S, Pan C, Hibiki T, Zhao J. Applications of nucleate boiling in renewable energy and thermal management and recent advances in modeling—a review. *Energy* 2024 Feb;289:129962. <https://doi.org/10.1016/j.energy.2023.129962>.
- [4] Emir T, Budakli M, Arik M. Pool boiling heat transfer: thermal performance for alternating and extended operational conditions. *Int J Heat Mass Tran* 2024 Jul;226:125454. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.125454>.
- [5] Ding T, Chen X, Li Z, Liu H, Zhu C, Zhao T, et al. A review of flow boiling heat transfer: theories, new methods and emerging applications. *Renew Sustain Energy Rev* 2025 Jun;215:115615. <https://doi.org/10.1016/j.rser.2025.115615>.
- [6] Lin L, Kedzierski MA. Review of low-GWP refrigerant pool boiling heat transfer on enhanced surfaces. *Int J Heat Mass Tran* 2019 Mar;131:1279–303. <https://doi.org/10.1016/j.ijheatmasstransfer.2018.11.142>.
- [7] Wang G, Kuang B, He Y, Liu P. Critical heat flux experiments for IVR-ERVC strategy under the pool boiling condition. In: Volume 7B: thermal-hydraulics and safety analysis. American Society of Mechanical Engineers; 2022. <https://doi.org/10.1115/1CONE29-93623>.
- [8] Zhang Z, Zhang Z, Wang C, Guo K, Tian W, Su G, et al. Heat pipe-cooled reactors: a comprehensive review of evolution, challenges, research status, and outlook. *Renew Sustain Energy Rev* 2025 May;213:115486. <https://doi.org/10.1016/j.rser.2025.115486>.
- [9] Shi J, Feng D, Chen Z. Experimental investigation on pool boiling heat transfer on untreated/super-hydrophilic metal foam under microgravity. *Int J Heat Mass Tran* 2020 Apr;151:119289. <https://doi.org/10.1016/j.ijheatmasstransfer.2019.119289>.
- [10] Chen G, Jia M, Zhang S, Tang Y, Wan Z. Pool boiling enhancement of novel interconnected microchannels with reentrant cavities for high-power electronics cooling. *Int J Heat Mass Tran* 2020 Aug;156:119836. <https://doi.org/10.1016/j.ijheatmasstransfer.2020.119836>.
- [11] Dadhich M, Prajapati OS. A brief review on factors affecting flow and pool boiling. *Renew Sustain Energy Rev* 2019 Sep;112:607–25. <https://doi.org/10.1016/j.rser.2019.06.016>.
- [12] Liu J, Tao L, Yang Q, Wang J. Recent advances in immersion cooling for thermal management of lithium-ion batteries. *Renew Sustain Energy Rev* 2026 Jan;226:116492. <https://doi.org/10.1016/j.rser.2025.116492>.
- [13] Xin Z, Tang W, Yao W, Wu Z. A review of thermal management of batteries with a focus on immersion cooling. *Renew Sustain Energy Rev* 2025 Jul;217:115751. <https://doi.org/10.1016/j.rser.2025.115751>.
- [14] Sun Y, Tang Y, Zhang S, Yuan W, Tang H. A review on fabrication and pool boiling enhancement of three-dimensional complex structures. *Renew Sustain Energy Rev* 2022 Jul;162:112437. <https://doi.org/10.1016/j.rser.2022.112437>.
- [15] Zhang C, Wang H, Huang Y, Zhang L, Chen Y. Immersion liquid cooling for electronics: materials, systems, applications and prospects. *Renew Sustain Energy Rev* 2025 Feb;208:114989. <https://doi.org/10.1016/j.rser.2024.114989>.
- [16] Lorfing D, Olives R, Falcoz Q, Guillot E, Le Men C, Ahmadi A. Design and performance of a new type of boiler using concentrated solar flux. *Energy Convers Manag* 2021 Dec;249:114835. <https://doi.org/10.1016/j.enconman.2021.114835>.
- [17] Xu Y, Liu M, Zhu J, Li H, Zhou W. Novel methods of oil fouling inhibition on surface of plate heat exchanger in simulated oilfield geothermal water. *Int J Heat Mass Tran* 2017 Oct;113:961–74. <https://doi.org/10.1016/j.ijheatmasstransfer.2017.05.107>.
- [18] Gupta SK, Nandy AK, Misra RD. Development of stable hybrid GNP–Pt–Al₂O₃ nanocomposite coatings via nanofluid pool boiling for advanced thermal management applications. *Arabian J Sci Eng* 2025 Jun;2. <https://doi.org/10.1007/s13369-025-10323-6>.
- [19] Jaikumar A, Kandlikar SG. Enhanced pool boiling for electronics cooling using porous fin tops on open microchannels with FC-87. *Appl Therm Eng* 2015 Dec;91:426–33. <https://doi.org/10.1016/j.applthermaleng.2015.08.043>.
- [20] Cardenas R, Narayanan V. Comparison of deionized water and FC-72 in pool and jet impingement boiling thermal management. *IEEE Trans Compon Packag Manuf Technol* 2012 Nov;2(11):1811–23. <https://doi.org/10.1109/TCPMT.2012.2210717>.
- [21] Ghaffari O, Grenier F, Morissette JF, Bolduc M, Jasmin S, Sylvestre J. Pool boiling experiment of dielectric liquids and numerical study for cooling a microprocessor. In: 2019 18th IEEE intersociety conference on thermal and thermomechanical phenomena in electronic systems (ITherm). IEEE; 2019. p. 540–5. <https://doi.org/10.1109/ITHERM.2019.8757380>.
- [22] Henry CD, Kim J. A study of the effects of heater size, subcooling, and gravity level on pool boiling heat transfer. *Int J Heat Fluid Flow* 2004 Apr;25(2):262–73. <https://doi.org/10.1016/j.ijheatfluidflow.2003.11.019>.

- [23] van Gils RW, Danilov D, Notten PHL, Speetjens MFM, Nijmeijer H. Battery thermal management by boiling heat-transfer. *Energy Convers Manag* 2014 Mar; 79:9–17. <https://doi.org/10.1016/j.enconman.2013.12.006>.
- [24] Al-Shetwi AQ, Atawi IE, El-Hameed MA, Abuelrub A. Digital twin technology for renewable energy, smart grids, energy storage and vehicle-To-grid integration: advancements, applications, key players, challenges and future perspectives in modernising sustainable grids. *IET Smart Grid* 2025 Jan 15;8(1). <https://doi.org/10.1049/stg2.70026>.
- [25] Phuangpornpitak N, Tia S. Opportunities and challenges of integrating renewable energy in smart grid system. *Energy Proc* 2013;34:282–90. <https://doi.org/10.1016/j.egypro.2013.06.756>.
- [26] Guo C, Luo F, Cai Z, Dong ZY. Integrated energy systems of data centers and smart grids: State-of-the-art and future opportunities. *Appl Energy* 2021 Nov;301: 117474. <https://doi.org/10.1016/j.apenergy.2021.117474>.
- [27] Patwary AK, Sayem MA, Hossain MA, Halim MdA. A review of energy storage systems (ESS) for integrating renewable energies in microgrids. *Control Syst Optimiz Lett* 2024 Apr 29;2(1):103–12. <https://doi.org/10.59247/csol.v2i1.68>.
- [28] Leong KC, Ho JY, Wong KK. A critical review of pool and flow boiling heat transfer of dielectric fluids on enhanced surfaces. *Appl Therm Eng* 2017 Feb;112: 999–1019. <https://doi.org/10.1016/j.applthermaleng.2016.10.138>.
- [29] Liang G, Mudawar I. Pool boiling critical heat flux (CHF) – part 1: review of mechanisms, models, and correlations. *Int J Heat Mass Tran* 2018 Feb;117: 1352–67. <https://doi.org/10.1016/j.ijheatmasstransfer.2017.09.134>.
- [30] Sajjad U, Sadeghianjahromi A, Ali HM, Wang CC. Enhanced pool boiling of dielectric and highly wetting liquids - a review on enhancement mechanisms. *Int Commun Heat Mass Tran* 2020 Dec;119:104950. <https://doi.org/10.1016/j.icheatmasstransfer.2020.104950>.
- [31] Emir T, Ourabi H, Budakli M, Arik M. Parametric effects on pool boiling heat transfer and critical heat flux: a critical review. *J Electron Packag* 2022 Dec 1;144 (4). <https://doi.org/10.1115/1.4054184>.
- [32] Volodin OA, Shvetsov DA, Serdyukov VS, Zhukov VI, Pavlenko AN. Enhanced boiling and evaporation of dielectric fluids on modified surfaces for immersion cooling of electronic components – a review. *Appl Therm Eng* 2025 Oct;277: 127088. <https://doi.org/10.1016/j.applthermaleng.2025.127088>.
- [33] Pereira JE, Souza RR, Moita AS. Experimental factors and techniques for pool boiling heat transfer enhancement: a critical review. *Heat Transfer* 2026 Feb 3. <https://doi.org/10.1002/hjt.70197>.
- [34] Xu C, Qian Z, Ren J. A comprehensive experimental investigation of additives to enhance pool boiling heat transfer of a non-azeotropic mixture. *Entropy* 2022 Oct 26;24(11):1534. <https://doi.org/10.3390/e24111534>.
- [35] Bar-Cohen A, Arik M, Ohadi M. Direct liquid cooling of high flux micro and nano electronic components. *Proc IEEE* 2006 Aug;94(8):1549–70. <https://doi.org/10.1109/JPROC.2006.879791>.
- [36] Incropera FP, DeWitt DP, Bergman TL, Lavine AS. *Fundamentals of heat and mass transfer*. New York: Wiley; 1996.
- [37] Bar-Cohen A, Simon TW. Wall superheat excursions in the boiling incipience of dielectric fluids. *Heat Transf Eng* 1988 Jun;9(3):19–31. <https://doi.org/10.1080/01457638808939668>.
- [38] El-Genk MS, Parker JL. Enhanced boiling of HFE-7100 dielectric liquid on porous graphite. *Energy Convers Manag* 2005 Sep;46(15–16):2455–81. <https://doi.org/10.1016/j.enconman.2004.11.012>.
- [39] Parker JL, El-Genk MS. Enhanced saturation and subcooled boiling of FC-72 dielectric liquid. *Int J Heat Mass Tran* 2005 Aug;48(18):3736–52. <https://doi.org/10.1016/j.ijheatmasstransfer.2005.03.011>.
- [40] El-Genk MS. Immersion cooling nucleate boiling of high power computer chips. *Energy Convers Manag* 2012 Jan;53(1):205–18. <https://doi.org/10.1016/j.enconman.2011.08.008>.
- [41] El-Genk MS, Suszko A. Effects of inclination angle and liquid subcooling on nucleate boiling on dimpled copper surfaces. *Int J Heat Mass Tran* 2016 Apr;95: 650–61. <https://doi.org/10.1016/j.ijheatmasstransfer.2015.12.048>.
- [42] Yu J, Chen Z, Utaka Y. Critical heat flux characteristics in pool boiling at low pressure for dielectric fluid Novec 7100. *Int J Heat Mass Tran* 2024 Nov;232: 125959. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.125959>.
- [43] El-Genk MS, Pourghasemi M. Subcooled boiling critical heat flux of HFE-7000 dielectric liquid on inclined rough Cu. *Int J Heat Mass Tran* 2021 Aug;175: 121354. <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121354>.
- [44] O'Connor JP, You SM, Chang JY. Gas-Saturated pool boiling heat transfer from smooth and microporous surfaces in FC-72. *J Heat Transfer* 1996 Aug 1;118(3): 662–7. <https://doi.org/10.1115/1.2822683>.
- [45] Liu ZW, Lin WW, Lee DJ, Peng XF. Pool boiling of FC-72 and HFE-7100. *J Heat Transfer* 2001 Apr 1;123(2):399–400. <https://doi.org/10.1115/1.1285892>.
- [46] El-Genk MS. Nucleate boiling enhancements on porous graphite and microporous and macro-finned copper surfaces. *Heat Transf Eng* 2012 Feb;33(3):175–204. <https://doi.org/10.1080/01457632.2011.589305>.
- [47] Al-Zareer M, Dincer I, Rosen MA. Novel thermal management system using boiling cooling for high-powered lithium-ion battery packs for hybrid electric vehicles. *J Power Sources* 2017 Sep;363:291–303. <https://doi.org/10.1016/j.jpowsour.2017.07.067>.
- [48] Pereira J, Souza R, Lima R, Moreira A, Moita A. An overview of the recent advances in pool boiling enhancement materials, structure, and devices. *Micromachines* 2024 Feb 17;15(2):281. <https://doi.org/10.3390/mi15020281>.
- [49] Kandlikar SG. In: Kandlikar SG, Shoji M, Dhir VK, editors. *Handbook of phase change: boiling and condensation*. Routledge; 2019. <https://doi.org/10.1201/9780203752654>.
- [50] Diaconu BM, Cruceru M, Anghelescu L. A critical review on heat transfer enhancement techniques in latent heat storage systems based on phase change materials. Passive and active techniques, system designs and optimization. *J Energy Storage* 2023 May;61:106830. <https://doi.org/10.1016/j.est.2023.106830>.
- [51] Qin S, Ji R, Miao C, Jin L, Yang C, Meng X. Review of enhancing boiling and condensation heat transfer: surface modification. *Renew Sustain Energy Rev* 2024 Jan;189:113882. <https://doi.org/10.1016/j.rser.2023.113882>.
- [52] Li W, Dai R, Zeng M, Wang Q. Review of two types of surface modification on pool boiling enhancement: passive and active. *Renew Sustain Energy Rev* 2020 Sep;130:109926. <https://doi.org/10.1016/j.rser.2020.109926>.
- [53] Chu H, Ji T, Yu X, Liu Z, Rui Z, Xu N. Advances in the application of machine learning to boiling heat transfer: a review. *Int J Heat Fluid Flow* 2024 Sep;108: 109477. <https://doi.org/10.1016/j.ijheatfluidflow.2024.109477>.
- [54] Wang Y, Wang N, Ren Q. Predicting surface heat flux on complex systems via Conv-LSTM. *Case Stud Therm Eng* 2022 May;33:101927. <https://doi.org/10.1016/j.csite.2022.101927>.
- [55] Al-Hindawi F, Soori T, Hu H, Siddiquee MdMR, Yoon H, Wu T, et al. A framework for generalizing critical heat flux detection models using unsupervised image-to-image translation. *Expert Syst Appl* 2023 Oct;227:120265. <https://doi.org/10.1016/j.eswa.2023.120265>.
- [56] Sajjad U, Mehdi S, Hussain I, Rehman T ur, Sultan M, Rashidi MM, et al. Physics driven interpretable deep learning-based insights into boiling crisis of smooth and roughened surfaces. *Alex Eng J* 2025 Mar;116:112–28. <https://doi.org/10.1016/j.aej.2024.10.103>.
- [57] Sajjad Uzair, Mehdi Sadaf, Zaidi Syed Shoaib Hassan, Ali Hafiz Muhammad. Machine learning driven metaheuristics to optimize the design-operational parameters for enhanced pool boiling CHF subject to a range of surface-liquid combinations. *Int J Heat Mass Tran* 2025;252:127510.
- [58] Pan L ming, wei Tan Z, Chen D qi, chang Xue L. Numerical investigation of vapor bubble condensation characteristics of subcooled flow boiling in vertical rectangular channel. *Nucl Eng Des* 2012 Jul;248:126–36. <https://doi.org/10.1016/j.nucengdes.2012.04.001>.
- [59] Euh D, Ozar B, Hibiki T, Ishii M, Song CH. Characteristics of bubble departure frequency in a low-pressure subcooled boiling flow. *J Nucl Sci Technol* 2010 Jul; 47(7):608–17. <https://doi.org/10.1080/18811248.2010.9720958>.
- [60] Bucci M, Buongiorno J, Bucci M. The not-so-subtle flaws of the force balance approach to predict the departure of bubbles in boiling heat transfer. *Phys Fluids* 2021 Jan 1;33(1). <https://doi.org/10.1063/5.0036956>.
- [61] Kon M, Kobayashi K, Watanabe M. Liquid temperature dependence of kinetic boundary condition at vapor-liquid interface. *Int J Heat Mass Tran* 2016 Aug;99: 317–26. <https://doi.org/10.1016/j.ijheatmasstransfer.2016.03.088>.
- [62] Pisano A. Physics in a vaporizer: saturated vapor pressure, heat of vaporization, and thermal expansion. In: *Physics for anesthetologists and intensivists*. Cham: Springer International Publishing; 2021. p. 173–84. https://doi.org/10.1007/978-3-030-72047-6_15.
- [63] Sielaff A, Dietl J, Herbert S, Stephan P. The influence of system pressure on bubble coalescence in nucleate boiling. *Heat Transf Eng* 2014 Mar 24;35(5): 420–9. <https://doi.org/10.1080/01457632.2013.830917>.
- [64] Sakashita H, Ono A. Boiling behaviors and critical heat flux on a horizontal plate in saturated pool boiling of water at high pressures. *Int J Heat Mass Tran* 2009 Jan;52(3–4):744–50. <https://doi.org/10.1016/j.ijheatmasstransfer.2008.06.040>.
- [65] Sakashita H. Bubble growth rates and nucleation site densities in saturated pool boiling of water at high pressures. *J Nucl Sci Technol* 2011 May;48(5):734–43. <https://doi.org/10.1080/18811248.2011.9711756>.
- [66] Dahariya S, Betz AR. High pressure pool boiling: mechanisms for heat transfer enhancement and comparison to existing models. *Int J Heat Mass Tran* 2019 Oct; 141:696–706. <https://doi.org/10.1016/j.ijheatmasstransfer.2019.07.016>.
- [67] Rainey KN, You SM, Lee S. Effect of pressure, subcooling, and dissolved gas on pool boiling heat transfer from microporous, square pin-finned surfaces in FC-72. *Int J Heat Mass Tran* 2003 Jan;46(1):23–35. [https://doi.org/10.1016/S0017-9310\(02\)00257-0](https://doi.org/10.1016/S0017-9310(02)00257-0).
- [68] Guan CK, Klausner JF, Mei R. A new mechanistic model for pool boiling CHF on horizontal surfaces. *Int J Heat Mass Tran* 2011 Aug;54(17–18):3960–9. <https://doi.org/10.1016/j.ijheatmasstransfer.2011.04.029>.
- [69] Dahariya S, Patel N, Egbo MK, Hwang G, Betz AR. High-pressure pool-boiling heat transfer enhancement mechanism on sintered-particle wick surface. *Front Mech Eng* 2020 Jan 22;5. <https://doi.org/10.3389/fmech.2019.00071>.
- [70] Zhou Z, Huang S, Huang H, Jiang M, Fang X. New correlation of the subcooled flow boiling heat transfer coefficient for hybrid micro/nanostructured surfaces with high heat flux incidence. *Phys Fluids* 2024 Feb 1;36(2). <https://doi.org/10.1063/5.0186992>.
- [71] Narayan S, Singh T, Srivastava A. Experiments on pool boiling regimes and bubble departure characteristics of single vapor bubble under subcooled bulk conditions. *Exp Therm Fluid Sci* 2020 Feb;111:109943. <https://doi.org/10.1016/j.expthermflusci.2019.109943>.
- [72] Lee J, Mudawar I. Fluid flow and heat transfer characteristics of low temperature two-phase micro-channel heat sinks – part 1: experimental methods and flow visualization results. *Int J Heat Mass Tran* 2008 Aug;51(17–18):4315–26. <https://doi.org/10.1016/j.ijheatmasstransfer.2008.02.012>.
- [73] Lee J, Mudawar I. Fluid flow and heat transfer characteristics of low temperature two-phase micro-channel heat sinks – part 2. Subcooled boiling pressure drop and heat transfer. *Int J Heat Mass Tran* 2008 Aug;51(17–18):4327–41. <https://doi.org/10.1016/j.ijheatmasstransfer.2008.02.013>.
- [74] Koeln JP, Alleyne AG. Optimal subcooling in vapor compression systems via extremum seeking control: theory and experiments. *Int J Refrig* 2014 Jul;43: 14–25. <https://doi.org/10.1016/j.jirefrig.2014.03.012>.

- [75] Jun S, Kim J, You SM, Kim HY. Effect of subcooling on pool boiling of water from sintered copper microporous coating at different orientations. *Sci Technol Nuclear Inst* 2018 Aug 13;2018:1–9. <https://doi.org/10.1155/2018/8623985>.
- [76] Watwe AA, Bar-Cohen A, McNeil A. Combined pressure and subcooling effects on pool boiling from a PPGA chip package. *J Electron Packag* 1997 Jun 1;119(2):95–105. <https://doi.org/10.1115/1.2792226>.
- [77] Rainey KN, You SM, Lee S. Effect of pressure, subcooling, and dissolved gas on pool boiling heat transfer from microporous surfaces in FC-72. *J Heat Transfer* 2003 Feb 1;125(1):75–83. <https://doi.org/10.1115/1.1527890>.
- [78] Sakurai A, Fukuda K. Mechanisms of subcooled pool boiling CHF depending on subcooling, pressure, and test heater configurations and surface conditions in liquids. In: *Heat transfer*, 3. ASME; 2002. p. 213–25. <https://doi.org/10.1115/IMECE2002-39066>.
- [79] Xiao L, Zhuang Y, Wu X, Yang J, Lu Y, Liu Y, et al. A review of pool-boiling processes based on bubble-dynamics parameters. *Appl Sci* 2023 Nov 3;13(21):12026. <https://doi.org/10.3390/app132112026>.
- [80] Schmidt WF, Volykhin KF, Khrapak AG, Illenberger E. Structure and mobility of positive and negative ions in non-polar liquids. *J Electrostat* 1999 Jun;47(1–2):83–95. [https://doi.org/10.1016/S0304-3886\(99\)00025-X](https://doi.org/10.1016/S0304-3886(99)00025-X).
- [81] Sudhakar T, Muniraj R, Jarin T, Sumathi S. Nanofluid-enhanced vegetable oil blends: a sustainable approach to breakthroughs in dielectric liquid insulation for electrical systems. *Biomass Convers Biorefinery* 2024 Apr 30. <https://doi.org/10.1007/s13399-024-05681-4>.
- [82] Clark FM. The development and application of synthetic liquid dielectrics. *Trans Electrochem Society* 1934;65(1):59. <https://doi.org/10.1149/1.3498057>.
- [83] Bluhm H, Frey W, Giese H, Hoppe P, Schultheiss C, Strassner R. Application of pulsed HV discharges to material fragmentation and recycling. *IEEE Trans Dielectr Electr Insul* 2000;7(5):625–36. <https://doi.org/10.1109/94.879358>.
- [84] Sprik M. Hydrogen bonding and the static dielectric constant in liquid water. *J Chem Phys* 1991 Nov 1;95(9):6762–9. <https://doi.org/10.1063/1.461515>.
- [85] Ballinger KWA, Michelson D, Przybyla JS. The study of the electrical dispersion of dielectric liquids in different gaseous environments. *J Electrostat* 1978 Sep;5:431–8. [https://doi.org/10.1016/0304-3886\(78\)90035-9](https://doi.org/10.1016/0304-3886(78)90035-9).
- [86] 3M. 3M™ fluorinert™ electronic liquid FC-72 product description, 1100; 2019. p. 1–4. 1100(September).
- [87] 3M. 3M novoc 649 engineered fluid. 2009. p. 8–11.
- [88] 3M. 3M™ Novoc™ 7100 engineered fluid. 2009. https://doi.org/10.1142/9789813275669_0001.
- [89] Ae A, Hydrofluoroether T. AsahiKlin™ AE3000. 1–4.
- [90] Wu X, Huang J, Zhuang Y, Liu Y, Yang J, Ouyang H, et al. Prediction models of saturated vapor pressure, saturated density, surface tension, viscosity and thermal conductivity of electronic fluoride liquids in two-phase liquid immersion cooling systems: a comprehensive review. *Appl Sci* 2023 Mar 26;13(7):4200. <https://doi.org/10.3390/app13074200>.
- [91] The Chemours Company FCL. Opteon™ 2P50 product information sheet. 2023. Report.
- [92] Inventec Performance Chemicals. THERMASOLV™ IM6 product data sheet. 2023. Report.
- [93] Mudawar I, Anderson TM. Parametric investigation into the effects of pressure, subcooling, surface augmentation and choice of coolant on pool boiling in the design of cooling systems for high-power-density electronic chips. *J Electron Packag* 1990 Dec 1;112(4):375–82. <https://doi.org/10.1115/1.2904392>.
- [94] Kutateladze Ls SS. Experimental investigation of liquid temperature influence on boiling regime variations. In: *Heat transfer problems under variable state of aggregation of matter*. GEI; 1953.
- [95] Ivey, H. J., and Morris DJ. On the relevance of the vapor liquid exchange mechanism for subcooled boiling heat transfer at higher pressure. Report SEW-R137. United Kin.
- [96] Zuber N. The hydrodynamic crisis in pool boiling of saturated and subcooled liquids. *Int. Dev. Heat Transfer* 1961;27:230–6.
- [97] Kanbur BB, Wu C, Fan S, Duan F. System-level experimental investigations of the direct immersion cooling data center units with thermodynamic and thermoeconomic assessments. *Energy* 2021 Feb;217:119373. <https://doi.org/10.1016/j.energy.2020.119373>.
- [98] Hameed MS, Khan AR, Mahdi AA. Modeling a general equation for pool boiling heat transfer. *Adv Chem Eng Sci* 2013;3(4):294–303. <https://doi.org/10.4236/aces.2013.34037>.
- [99] Baldwin M, Ghavami A, Ghiaasiaan SM, Majumdar A. Pool boiling in liquid hydrogen, liquid methane and liquid oxygen: a review of available data and predictive tools. *Cryogenics* 2021 Apr;115:103240. <https://doi.org/10.1016/j.cryogenics.2020.103240>.
- [100] Rohsenow WM, Griffith P, Office THE, Naval OF. Technical report no. 6 correlation of maximum heat flux data for boiling of saturated liquids. Massachusetts Institute of Technology; 1955.
- [101] Zuber N. Hydrodynamic aspects of boiling heat transfer (Thesis). In: Report; 1959 Jun. <https://doi.org/10.2172/4175511>. Oak Ridge, TN.
- [102] Lienhard JH, Dhir VK. Hydrodynamic prediction of peak pool-boiling heat fluxes from finite bodies. *J Heat Transfer* 1973 May 1;95(2):152–8. <https://doi.org/10.1115/1.3450013>.
- [103] Bewilogua L, Knöner R, Vinzelberg H. Heat transfer in cryogenic liquids under pressure. *Cryogenics* 1975 Mar;15(3):121–5. [https://doi.org/10.1016/0011-2275\(75\)90178-2](https://doi.org/10.1016/0011-2275(75)90178-2).
- [104] Brentari EG, Giarratano Patricia J. RVS. Boiling heat transfer for oxygen, nitrogen, hydrogen, and helium. 317. US National Bureau of Standards; 1965.
- [105] Haramura Y, Katto Y. A new hydrodynamic model of critical heat flux, applicable widely to both pool and forced convection boiling on submerged bodies in saturated liquids. *Int J Heat Mass Tran* 1983 Mar;26(3):389–99. [https://doi.org/10.1016/0017-9310\(83\)90043-1](https://doi.org/10.1016/0017-9310(83)90043-1).
- [106] Galloway JE, Mudawar I. CHF mechanism in flow boiling from a short heated wall—II. Examination of near-wall conditions with the aid of photomicrography and high-speed video imaging. *Int J Heat Mass Tran* 1993 Jul;36(10):2511–26. [https://doi.org/10.1016/S0017-9310\(05\)80190-5](https://doi.org/10.1016/S0017-9310(05)80190-5).
- [107] Galloway JE, Mudawar I. CHF mechanism in flow boiling from a short heated wall—II. Theoretical CHF model. *Int J Heat Mass Tran* 1993 Jul;36(10):2527–40. [https://doi.org/10.1016/S0017-9310\(05\)80191-7](https://doi.org/10.1016/S0017-9310(05)80191-7).
- [108] Howard AH, Mudawar I. Orientation effects on pool boiling critical heat flux (CHF) and modeling of CHF for near-vertical surfaces. *Int J Heat Mass Tran* 1999 May;42(9):1665–88. [https://doi.org/10.1016/S0017-9310\(98\)00233-6](https://doi.org/10.1016/S0017-9310(98)00233-6).
- [109] Deev VI, Keilin VE, Kovalev IA, Kondratenko AK, Petrovichev VI. Nucleate and film pool boiling heat transfer to saturated liquid helium. *Cryogenics* 1977 Oct;17(10):557–62. [https://doi.org/10.1016/0011-2275\(77\)90259-4](https://doi.org/10.1016/0011-2275(77)90259-4).
- [110] Wang L, Li Y, Zhang F, Xie F, Ma Y. Correlations for calculating heat transfer of hydrogen pool boiling. *Int J Hydrogen Energy* 2016 Oct;41(38):17118–31. <https://doi.org/10.1016/j.ijhydene.2016.06.254>.
- [111] Bailey W, Young E, Beduz C, Yang Y. Pool boiling study on candidature of pentane, methanol and water for near room temperature cooling. In: *Thermal and thermomechanical proceedings 10th intersociety conference on phenomena in electronics systems*, 2006. ITherm 2006. IEEE. p. 599–603. doi:10.1109/ITHERM.2006.1645400.
- [112] Sakashita H. Critical heat flux on a vertical surface in saturated pool boiling at high pressures. *J Therm Sci Technol* 2016;11(2):JTST0020. <https://doi.org/10.1299/jtst.2016.jtst0020>.
- [113] YuA Kirichenko, Chernyakov PS. Determination of the first critical thermal flux on flat heaters. *J Eng Phys* 1971 Jun;20(6):699–703. <https://doi.org/10.1007/BF01122589>.
- [114] Ahmad F, Kim S, Meyer M, Hartwig J, Mudawar I. Saturated nucleate pool boiling of cryogenic fluids: review of databases, assessment of existing models and correlations, and development of new universal correlation. *Int J Heat Mass Tran* 2024 Oct;231:125807. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.125807>.
- [115] Kruzhilin GN. Free-convection transfer of heat from a horizontal plate and boiling liquid. *Doklady AN SSSR (reports of the USSR Academy of Sciences)* 1947;58(8):1657–60.
- [116] McNelly MJ. A correlation of rates of heat transfer to nucleate boiling of liquids. *J Imperial College Chem Eng Soc* 1953;7:18–34.
- [117] Forster HK, Zuber N. Dynamics of vapor bubbles and boiling heat transfer. *AIChE J* 1955 Dec 17;1(4):531–5. <https://doi.org/10.1002/aic.690010425>.
- [118] Kutateladze SS. Heat transfer in condensation and boiling. *US Atomic Energy Commission, Tech Inform Service* 1959;3770.
- [119] Cooper MG. Saturation nucleate pool boiling - a simple correlation. In: *First UK, editor. National conference on heat transfer*. Elsevier; 1984. p. 785–93. <https://doi.org/10.1016/B978-0-85295-175-0.50013-8>.
- [120] Kim J, Benton JF. Highly subcooled pool boiling heat transfer at various gravity levels. *Int J Heat Fluid Flow* 2002 Aug;23(4):497–508. [https://doi.org/10.1016/S0142-727X\(02\)00139-X](https://doi.org/10.1016/S0142-727X(02)00139-X).
- [121] Raj R, Kim J, McQuillen J. Pool boiling heat transfer on the international space station: experimental results and model verification. *J Heat Transfer* 2012 Oct 1;134(10). <https://doi.org/10.1115/1.4006846>.
- [122] Wang X, Du Y, Tang Y, Liu L, Liu B, Zhang B, et al. Heater size and gravity level effects on nucleate pool boiling heat transfer under controlled wall heat flux condition. *Exp Therm Fluid Sci* 2022 Jun;134:110603. <https://doi.org/10.1016/j.expthermflusci.2022.110603>.
- [123] Raj R, Kim J, McQuillen J. Gravity scaling parameter for pool boiling heat transfer. *J Heat Transfer* 2010 Sep 1;132(9). <https://doi.org/10.1115/1.4001632>.
- [124] Di Marco P, Grassi W. Effect of force fields on pool boiling flow patterns in normal and reduced gravity. *Heat Mass Tran* 2009 May 11;45(7):959–66. <https://doi.org/10.1007/s00231-007-0328-6>.
- [125] Zhao JF, Li J, Yan N, Wang SF. Bubble behavior and heat transfer in quasi-steady pool boiling in microgravity. *Microgravity Sci Technol* 2009 Aug 7;21(S1):175–83. <https://doi.org/10.1007/s12217-009-9151-7>.
- [126] Raj R, Kim J. Heater size and gravity based pool boiling regime map: transition criteria between buoyancy and surface tension dominated boiling. *J Heat Transfer* 2010 Sep 1;132(9). <https://doi.org/10.1115/1.4001635>.
- [127] Garivalis AI, Liu B, Wei J, Di Marco P. Enhancing pool boiling in microgravity using micro-fins and an electrostatic field under different liquid subcooling. *Exp Therm Fluid Sci* 2024 Jun;155:111179. <https://doi.org/10.1016/j.expthermflusci.2024.111179>.
- [128] Parker JL, El-Genk MS. Enhanced saturation and subcooled boiling of FC-72 dielectric liquid. *Int J Heat Mass Tran* 2005;48(18):3736–52. <https://doi.org/10.1016/j.ijheatmasstransfer.2005.03.011>.
- [129] Kandlikar SG. History, advances, and challenges in liquid flow and flow boiling heat transfer in microchannels: a critical review. *J Heat Transfer* 2012 Mar 1;134(3). <https://doi.org/10.1115/1.4005126>.
- [130] El-Genk MS, Parker JL. Pool boiling in saturated and subcooled HFE-7100 dielectric fluid from a porous graphite surface. In: *The ninth intersociety conference on thermal and thermomechanical phenomena in electronic systems* (IEEE cat. No.04CH37543). IEEE. p. 655–662. doi:10.1109/ITHERM.2004.1319238.
- [131] Dedov AV, Filippov MD. Study of boiling heat transfer on modified surfaces of dielectric liquid Novoc 7100. *Heat Mass Tran* 2025 Apr 9;61(4):38. <https://doi.org/10.1007/s00231-025-03559-3>.

- [132] Zhou J, Qi B, Wei J. Critical heat flux on heterogeneous fractal surfaces with micro-pin-fins in pool boiling part I: the effects of distribution and subcooling. *Int J Heat Mass Tran* 2019;136:1338–48. <https://doi.org/10.1016/j.ijheatmasstransfer.2019.01.089>.
- [133] Sathyamurthi V, Ahn HS, Banerjee D, Lau SC. Subcooled pool boiling experiments on horizontal heaters coated with carbon nanotubes. *J Heat Transfer* 2009 Jul 1; 131(7). <https://doi.org/10.1115/1.3000595>.
- [134] Wei JJ, Guo LJ, Honda H. Experimental study of boiling phenomena and heat transfer performances of FC-72 over micro-pin-finned silicon chips. *Heat Mass Tran* 2005 Jun 31;41(8):744–55. <https://doi.org/10.1007/s00231-005-0633-x>.
- [135] Ono A, Sakashita H. Liquid–vapor structure near heating surface at high heat flux in subcooled pool boiling. *Int J Heat Mass Tran* 2007 Aug;50(17–18):3481–9. <https://doi.org/10.1016/j.ijheatmasstransfer.2007.01.026>.
- [136] Fan X, Gu S, Lei J, Luo G, Meng F, Wu L, et al. Experimental and analytical study on the influence of saturation pressure and surface roughness on pool boiling CHF of HFE-7100. In: Wu R, editor. *International journal of chemical engineering*. 2022; 2022 Jun 29. p. 1–18. <https://doi.org/10.1155/2022/4875208>.
- [137] Gao L, Bai M, Lyu J, Li Y, Li Y, Lv X. Subcooled pool boiling performance on conical micro-pillars structures surfaces. *Exp Therm Fluid Sci* 2023 Oct;148: 110993. <https://doi.org/10.1016/j.expthermflusci.2023.110993>.
- [138] Wu F, Umehara Y, Etoh A, Mori S. Effects of pore size distribution in a honeycomb porous plate on the onset of nucleate boiling in AE-3000. *Int J Heat Mass Tran* 2025 Aug;246:127066. <https://doi.org/10.1016/j.ijheatmasstransfer.2025.127066>.
- [139] Kano I. Pool boiling enhancement by electrohydrodynamic force and diamond coated surfaces. *J Heat Transfer* 2015 Sep 1;137(9). <https://doi.org/10.1115/1.4030217>.
- [140] Walunj A, Sathyabhama A. Transient CHF enhancement in high pressure pool boiling on rough surface. *Chem Eng Process Process Intensif* 2018 May;127: 145–58. <https://doi.org/10.1016/j.ccep.2018.03.025>.
- [141] Kandlikar SG. A theoretical model to predict pool boiling CHF incorporating effects of contact angle and orientation. *J Heat Transfer* 2001 Dec 1;123(6): 1071–9. <https://doi.org/10.1115/1.1409265>.
- [142] Arik M, Bar-Cohen A. Pool boiling of perfluorocarbon mixtures on silicon surfaces. *Int J Heat Mass Tran* 2010 Nov;53(23–24):5596–604. <https://doi.org/10.1016/j.ijheatmasstransfer.2010.06.034>.
- [143] Fariñas Alvarino P, Sánchez Simón ML, dos Santos Guzella M, Amado Paz JM, Sáiz Jabardo JM, Cabezas Gómez L. Experimental investigation of the CHF of HFE-7100 under pool boiling conditions on differently roughened surfaces. *Int J Heat Mass Tran* 2019 Aug;139:269–79. <https://doi.org/10.1016/j.ijheatmasstransfer.2019.04.142>.
- [144] Rohsenow WM. A method of correlating heat-transfer data for surface boiling of liquids. *J Fluids Eng* 1952 Aug 1;74(6):969–75. <https://doi.org/10.1115/1.4015984>.
- [145] Kutateladze SS. Heat transfer in condensation and boiling, 3770. US Atomic Energy Commission; 1959. Technical Information Service.
- [146] Cooper MG. Heat flow rates in saturated nucleate pool Boiling-A wide-ranging examination using reduced properties. 1984. p. 157–239. [https://doi.org/10.1016/S0065-2717\(08\)70205-3](https://doi.org/10.1016/S0065-2717(08)70205-3).
- [147] Pioro I I. Experimental evaluation of constants for the Rohsenow pool boiling correlation. *Int J Heat Mass Tran* 1999 Jun;42(11):2003–13. [https://doi.org/10.1016/S0017-9310\(98\)00294-4](https://doi.org/10.1016/S0017-9310(98)00294-4).
- [148] Kutateladze SS. Boiling heat transfer. *Int J Heat Mass Tran* 1961 Dec;4:31–45. [https://doi.org/10.1016/0017-9310\(61\)90059-X](https://doi.org/10.1016/0017-9310(61)90059-X).
- [149] Choon NK, Chakraborty A, Aye SM, Xiaolin W. New pool boiling data for water with copper-foam metal at sub-atmospheric pressures: experiments and correlation. *Appl Therm Eng* 2006 Aug;26(11–12):1286–90. <https://doi.org/10.1016/j.applthermaleng.2005.10.028>.
- [150] Khezripour Z, Etesami N, Karshenas HR. Boiling curve of subcooled flow boiling of DI water from forced convection region until reaching CHF under different operation parameters in a vertical tube. *J Therm Anal Calorim* 2024 Nov 31;149 (21):12439–53. <https://doi.org/10.1007/s10973-024-13572-z>.
- [151] Galicia ES, Otomo Y, Saiwai T, Takita K, Orito K, Enoki K. Subcooled flow boiling heat flux enhancement using high porosity sintered fiber. *Appl Sci* 2021 Jun 24; 11(13):5883. <https://doi.org/10.3390/app11135883>.
- [152] Wu C. Feasibility study of immersion cooling for data center. Nanyang Technological University; 2025. <https://doi.org/10.32657/10356/204506>.
- [153] Investing.com. Real-time SGD/KRW exchange rate. 2025.
- [154] Bejan A, Tsatsaronis G, Moran MJ. Thermal design and optimization. John Wiley & Sons; 1995.
- [155] Timmerhaus KD, West RE. Plant design and economics for chemical engineers. fifth ed. Boston: McGraw-Hill; 2003.
- [156] Smith R. Chemical process: design and integration. New York: Wiley; 2014.
- [157] Woods DR. Rules of thumb in engineering practice. New York: Wiley; 2007.
- [158] GlobalPetrolPrices.com. South Korea electricity prices; 2025.
- [159] Climate transparency. Climate transparency report 2022: south Korea. 2022.
- [160] International Emissions Trading Association. Korea business brief. 2025. Report.